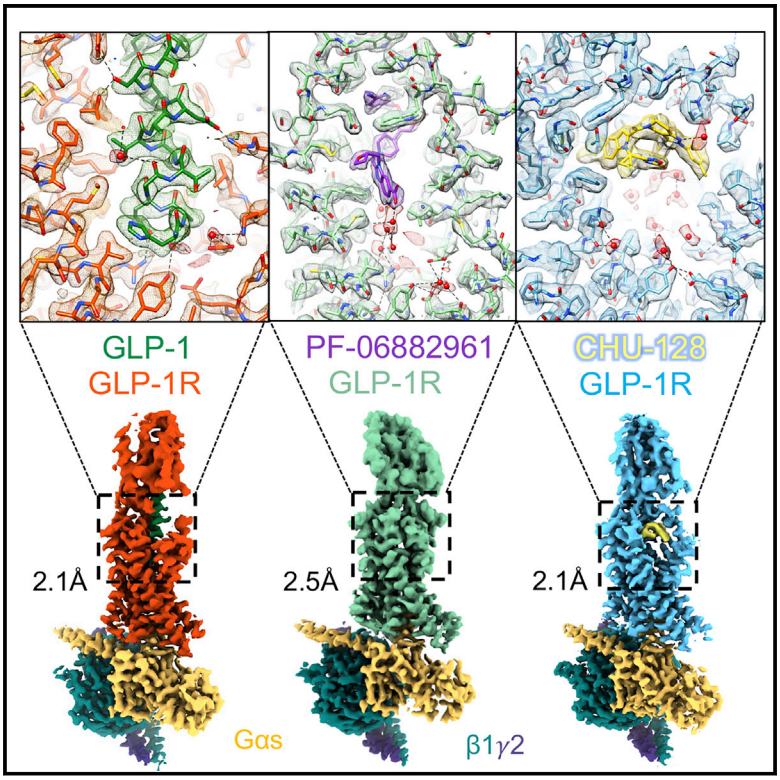
Molecular Cell

Article

Differential GLP-1R Binding and Activation by Peptide and Non-peptide Agonists

Graphical Abstract



Highlights

- 2.1-, 2.1-, and 2.5-Å GLP-1R:Gs structures bound to GLP-1 and non-peptide agonists
- Critical water networks stabilize peptide and small-molecule agonist binding
- PF 06882961, but not CHU-128, closely mimics the GLP-1 signaling profile
- Insight into how select non-peptide agonists mimic receptor activation by GLP-1

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In Brief

Zhang et al. show that the non-peptide GLP-1R agonist PF 06882961, but not CHU-128, was able to closely mimic the *in vitro* pharmacological fingerprint of GLP-1. These pharmacological activities could be rationalized from high-resolution structures that provide a novel template for the design of new agonists targeting the GLP-1R.

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Differential GLP-1R Binding and Activation by Peptide and Non-peptide Agonists

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SUMMARY

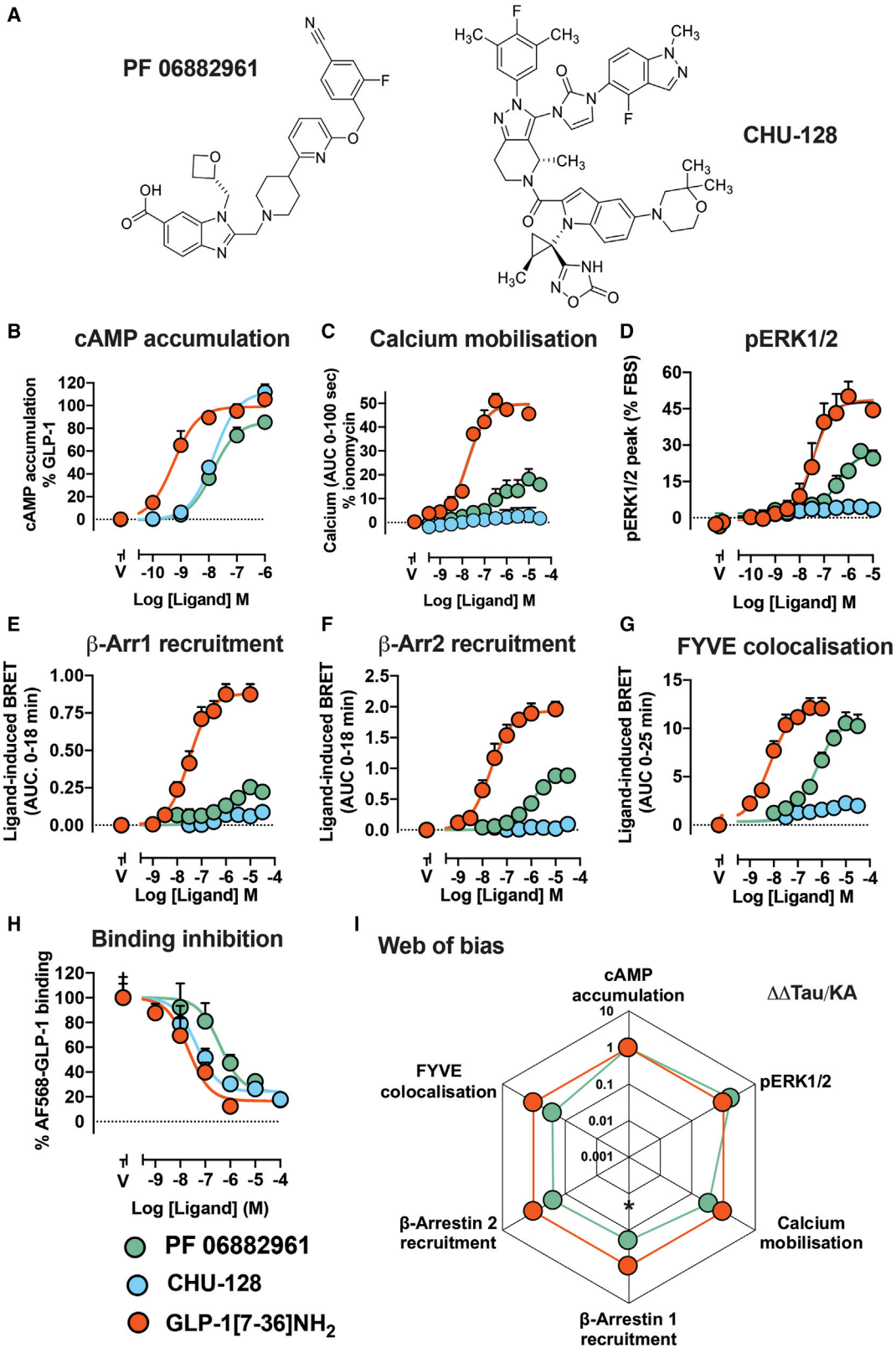
Peptide drugs targeting class B1 G-protein-coupled receptors (GPCRs) can treat multiple diseases; however, there remains substantial interest in the development of orally delivered non-peptide drugs. Here, we reveal unexpected overlap between signaling and regulation of the glucagon-like peptide-1 (GLP-1) receptor by the non-peptide agonist PF 06882961 and GLP-1 that was not observed for another compound, CHU-128. Compounds from these patent series, including PF 06882961, are currently in clinical trials for treatment of type 2 diabetes. High-resolution cryoelectron microscopy (cryo-EM) structures reveal that the binding sites for PF 06882961 and GLP-1 substantially overlap, whereas CHU-128 adopts a unique binding mode with a more open receptor conformation at the extracellular face. Structural differences involving extensive water-mediated hydrogen bond networks could be correlated to functional data to understand how PF 06882961, but not CHU-128, can closely mimic the pharmacological properties of GLP-1. These findings will facilitate rational structure-based discovery of non-peptide agonists targeting class B GPCRs.

INTRODUCTION

Class B1 G-protein-coupled receptors (GPCRs) are activated by peptide hormones/neuropeptides and regulate physiological functions, including glycemic control, satiety, neuroprotection, bone turnover, and cardiovascular tone (Culhane et al., 2015; Pal et al., 2012). These receptors couple to the stimulatory protein G_s to promote cyclic AMP (cAMP) production; however, they can activate a range of signaling pathways and are highly susceptible to biased agonism (Wootten et al., 2017). Numerous active state structures of class B1 GPCRs coupled to G_s have been determined using single-particle cryoelectron microscopy (cryo-EM) (Liang et al., 2017, Liang et al., 2018a, Liang et al., 2018b, Liang et al., 2020a, Liang et al., 2020b; Ma et al., 2020; Qiao et al., 2020; Zhang et al., 2017; Zhao et al., 2019), revealing common peptide-binding pockets and a conserved G_s-binding site within the receptor transmembrane (TM) bundle.

The glucagon-like peptide-1 receptor (GLP-1R) is a class B1 GPCR that is the target of multiple clinically approved peptide drugs for the treatment of type 2 diabetes, with select drugs also approved for obesity (Brown et al., 2018; Nauck and Meier, 2019). These peptides have been engineered to enhance pharmacokinetic profiles over the rapidly cleared native GLP-1 peptides while maintaining potency at the GLP-1R. Nonetheless, there are differences in their clinical efficacies (Brown et al., 2018; Grieco et al., 2019; Nauck and Meier, 2019). To date, the structures of active-state GLP-1R:G_s complexes bound to either GLP-1 or a G protein biased agonist, exendin-P5 (ExP5), have been determined (Liang et al., 2018b; Zhang et al., 2017). In addition, a full-length GLP-1R structure bound by an extracellular antibody, in the absence of a peptide ligand, was reported (Wu et al., 2020). These structures provide critical information regarding GLP-1R activation by peptides and have provided initial insights into the molecular basis of differential efficacy and biased agonism.

All GLP-1 peptide analogs require parenteral administration, except for semaglutide, which is also approved in oral formulation. Moreover, peptides are expensive to produce and induce



(legend on next page)

nausea and other gastrointestinal problems that reduce patient compliance (Pearson et al., 2019). Consequently, orally available small molecules that can also broadly mimic the actions of GLP-1 are of major interest. Several non-peptide agonists have entered clinical trials, including PF 06882961 (Pfizer), TTP-273 (vTv Therapeutics/Huadong Medicine), and OWL-833 (Chugai/Eli Lilly). TT-OAD2, an analog reported in the patent series that includes TTP-273, displays unique kinetic and signaling properties and has a distinct binding mode relative to GLP-1 (Zhao et al., 2020). Little is known regarding the binding mode, activation, and signaling profiles of PF 06882961, OWL-833, or related compounds. These molecules have greater activity at the GLP-1R than TT-OAD2, suggesting greater promise in the clinic. Indeed, recent phase one clinical data showed robust reductions in fasting plasma glucose and body weight in adults with type 2 diabetes following 28 days of treatment with PF 06882961 (Saxena et al., 2020).

Although the structure of PF 06882961 is available, OWL-833 has not been disclosed, albeit the patent series of GLP-1R small-molecule agonists filed by Chugai is available (W02018056453/US 20190225604A1). Here, we revealed that PF 06882961 has a remarkably similar overall signaling and regulatory profile to GLP-1, whereas CHU-128 (from the Chugai patent series) exhibited a distinct profile that is strongly biased toward G_s-mediated cAMP production relative to the spectrum of responses elicited by GLP-1. High-resolution structural information on the active GLP-1R-G_s signaling complex in the presence of GLP-1, PF 06882961, and CHU-128 provides unique mechanistic insights into the divergent binding modes of each agonist and can explain how PF 06882961, but not CHU-128, is able to mimic the actions of GLP-1.

RESULTS AND DISCUSSION

PF 06882961 and CHU-128 Have Different Pharmacological Profiles

The GLP-1R is pleiotropically coupled to multiple transducer and regulatory proteins (Graaf et al., 2016). As such, we assessed the signaling profiles of GLP-1, PF 06882961, and CHU-128 using time-resolved assays of well-studied downstream signaling and regulatory events linked to GLP-1R activation: cAMP production; pERK1/2; intracellular calcium mobilization; and β-arrestin-1 and β-arrestin-2 recruitment. We also monitored ligand-induced G_s conformational changes (associated with G_s activation and cAMP signaling) and receptor trafficking into FYVE-containing early endosomes, where continued peptide-mediated signaling can contribute to cellular response (Figures 1 and S1).

PF 06882961 and CHU-128 exhibited equivalent potency and were full agonists for cAMP production, with ~30-fold lower potency than GLP-1 (Figure 1B). This was correlated with similar potencies for the two non-peptides in inducing conformational changes in G_s (Figure S1A); however, CHU-128 was inactive in all other assays at concentrations up to 10 μM (Figures 1C–1G and S1). In marked contrast, PF 06882961 was active in all assays (30- to 100-fold lower potency than GLP-1), displaying similar maximal internalization to GLP-1 but only partial agonism in pERK1/2, calcium mobilization, and β-arrestin recruitment (Figures 1B–1G and S1). Quantification of responses using operational modeling revealed subtle biased agonism of PF 06882961, relative to GLP-1 that was only significantly different for β-arrestin-1 recruitment relative to cAMP (Figure 1I). However, although PF 06882961 had a similar kinetic profile to GLP-1 in inducing G_s conformational change, calcium signaling, and pERK1/2, it was slower at inducing β-arrestin recruitment and internalization (Figure S1). Nonetheless, given the major divergence in the chemical properties (Figure 1A), overall PF 06882961 exhibited a remarkably similar signaling and regulatory profile to GLP-1 (Figure 1I). Indeed, many peptide agonists, including the clinically approved peptides, exendin-4 and liraglutide, and endogenous oxyntomodulin, demonstrate greater biased agonism relative to GLP-1 across the assessed pathways than PF 06882961 (Fletcher et al., 2018; Hager et al., 2016; Koole et al., 2010).

Due to the lack of response in the majority of assays, bias could not be quantified for CHU-128, but, as it had an equivalent response to PF 06882961 in cAMP production, we can infer that CHU-128 is heavily biased toward this pathway, and away from the other pathways, when compared to GLP-1 and PF 06882961. Interestingly, assessment of binding properties revealed CHU-128 had ~10-fold higher affinity than PF 06882961 (Figures 1H and S1G), suggesting that, although the two agonists have similar concentration response profiles in cAMP accumulation (Figure 1B), PF 06882961 has higher efficacy for this pathway.

Intriguingly, CHU-128 has a similar signaling and regulatory profile to TT-OAD2, where neither agonist promoted β-arrestin recruitment or internalization and both were also very poor agonists for pERK1/2 and calcium mobilization. Nonetheless, CHU-128 was a more efficient agonist for cAMP production, with higher potency and maximal responses than TT-OAD2 (Figure 1; Zhao et al., 2020).

Collectively, these data suggest PF 06882961 is likely to have relatively similar pharmacodynamics properties to GLP-1 *in vivo*, although CHU-128 and TT-OAD2 may exhibit greater similarity in

Figure 1. Pharmacological Profiles of GLP-1, PF 06882961, and CHU-128

(A) PF 06882961 and CHU-128.

(B–G) Concentration response curves showing the signaling profiles in cAMP accumulation (B), calcium mobilization (C), pERK1/2 (D), β-arrestin-1 recruitment (E), β-arrestin-2 recruitment (F), and GLP-1R internalization (measured by ligand-induced colocalization with the early endosome marker, FYVE (G). (C)–(G) were plotted using the area under the curve (AUC) analysis applied to data from Figure S1.

(H) Inhibition of BRET between NLuc-GLP-1R and AF568-GLP-1 by PF 06882961 and CHU-128. Data were derived from the last time point in Figure S1G.

(I) "Web of bias" plots $\Delta\Delta\tau/K_A$ values on a logarithmic scale showing signaling profile of PF 06882961 relative to GLP-1 and the cAMP pathway. τ/K_A was calculated using the operational model of agonism, applied to data in (B)–(G). * indicates statistically significant bias ($p < 0.05$) relative to GLP-1 as determined by a two-way ANOVA followed by a Sidak's multiple comparisons test, comparing each pathway to cAMP and PF 06882961 to GLP-1. All data are the means + SEM of 5 independent experiments performed in duplicate or triplicate, with the exception of (H) (3 independent experiments).

See also Figure S1.

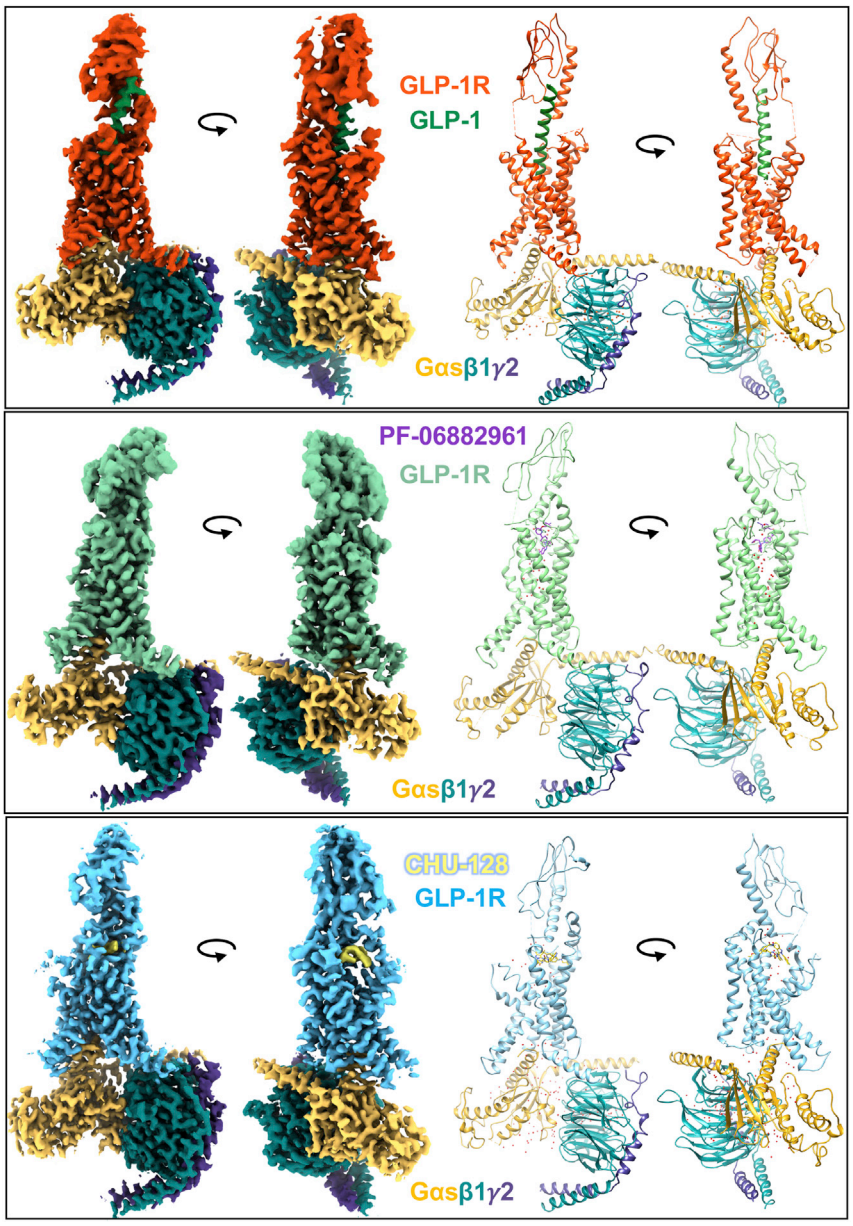


Figure 2. cryo-EM Structures of GLP-1R:G_s Complexes

Top: GLP-1 bound; middle: PF 06882961 bound; bottom: CHU-128 bound; orthogonal views of the cryo-EM maps and backbone models built into the maps in ribbon format. Coloring denotes the protein segments as highlighted. See also Figures 3, S2, S3, and S7 and Videos S1, S3, and S4.

their pharmacodynamic profiles (albeit with markedly different kinetics). However, these data also raise critical mechanistic and structural questions on how the different small-molecule agonists engage and activate the GLP-1R and in particular with respect to how PF 06882961 can mimic key properties of the native peptide.

Determination of High-Resolution Structures of Agonist-Bound GLP-1R:G_s Complexes

To understand agonist engagement with the GLP-1R, we determined human GLP-1R:G_s structures bound to GLP-1, PF 06882961, or CHU-128 (Figures 2 and S2). Complexes were induced by the addition of 10 μM GLP-1 or 50 μM of small molecules prior to solubilization and purification. Purified complexes were resolved as monodispersed peaks on size-exclusion chromatography (SEC) and contained all the expected components (Figure S2). Vitrified complexes were imaged by cryo-EM yielding 2D class averages that were reconstructed into 3D consensus density maps with global resolutions of 2.1 Å for GLP-1, 2.5 Å for PF 06882961, and 2.1 Å for CHU-128 at gold standard Fourier shell correlation (FSC) 0.143 (Figure S2; Table S1). The α-helical domain (AHD) of the $G\alpha_s$ subunit was poorly resolved in all consensus maps and was masked out during the final refinement. To improve the resolution of domains with higher mobility and thus lower local resolution in consensus maps, localized 3D refinements focused on the full receptor, the receptor extracellular domain (ECD), or the G protein were also performed (Figure S2) and enabled robust molecular modeling within the receptor ligand binding pockets and the ECD for each complex.

Using a combination of the consensus, full-receptor-refined, and ECD-refined maps of the GLP-1-bound complex, atomic level modeling of the entire peptide, almost the full receptor, and G protein, excepting the $G\alpha_s$ AHD, was performed (Figures S2–S4). Within the PF-06882961- and CHU-128-bound GLP-1R cryo-EM maps, the GLP-1R TM domain was well resolved allowing unambiguous modeling of both ligands and the majority of receptor residues, including the binding sites and the majority of the loop regions, with the exception of intracellular loop (ICL) 3, where only the backbone was modeled, and the TM1/ECD linkers, where S129–S134 was not modeled (Figures 2 and S3). Extracellular loop (ECL) 3 between H374 and T377 was also poorly resolved and not modeled for the CHU-128-bound receptor. The GLP-1R ECD in the non-peptide bound complexes was better resolved than in the GLP-1-bound map (Figure S2), enabling unambiguous modeling of the backbone and the majority of side chains using the receptor-refined maps.

The GLP-1 Peptide Forms Extensive Interactions with the GLP-1R TM Domain and Structural Waters

Structural understanding of the mechanism through which PF 06882961, but not CHU-128, can mimic the pharmacology of

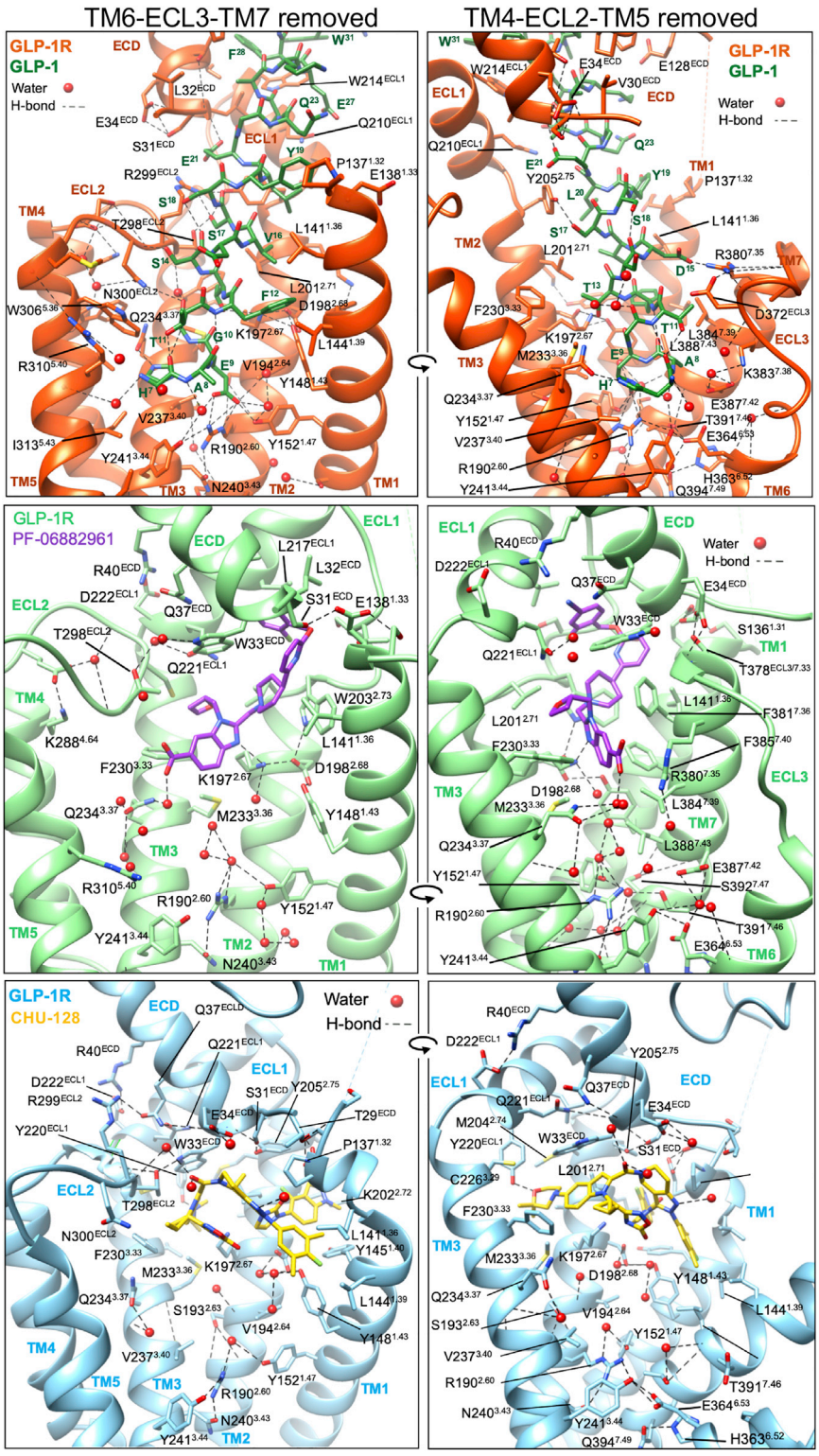
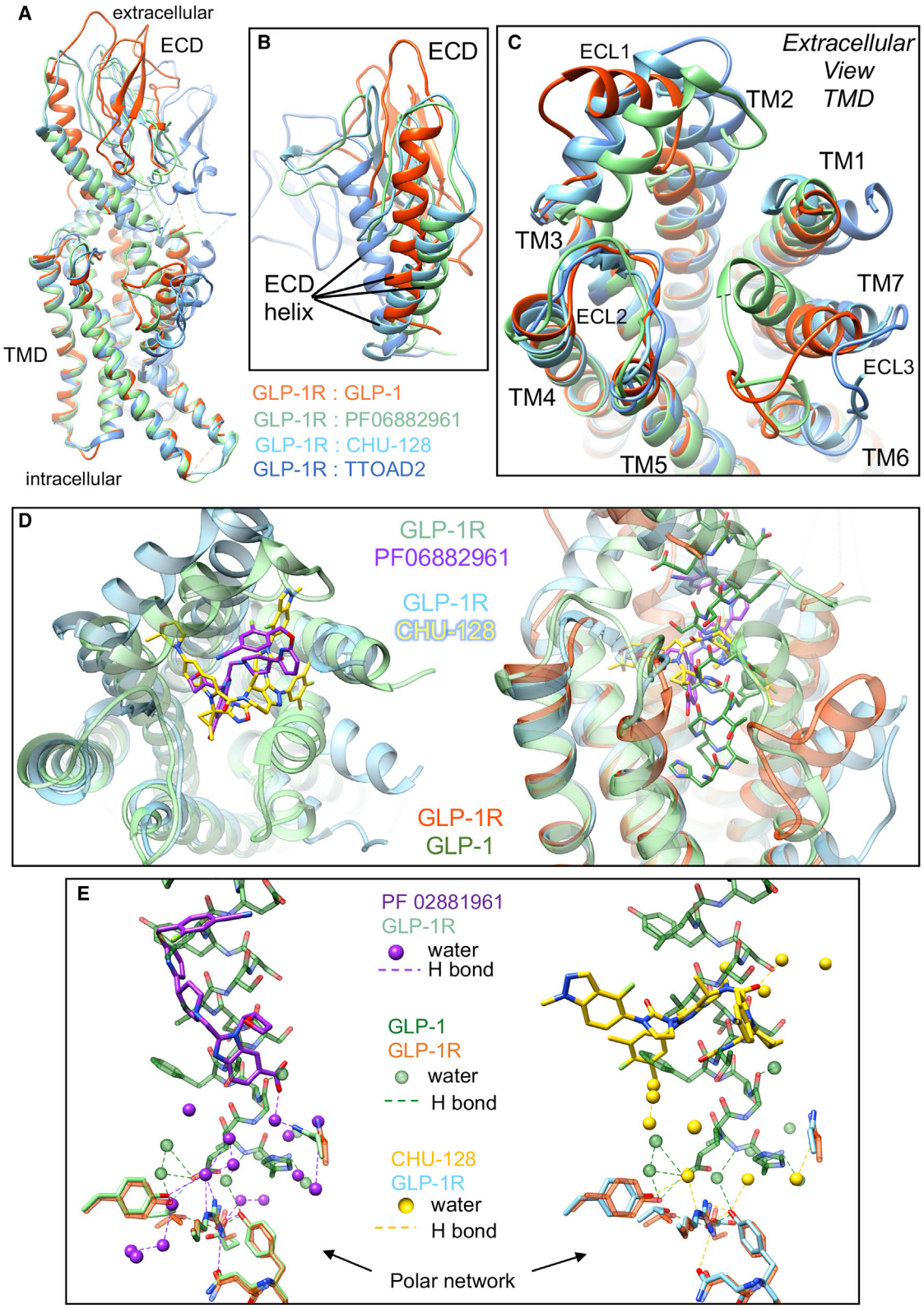


Figure 3. Interactions of GLP-1 and Non-peptidic Agonists within the GLP-1R TM Binding Cavity

Top: GLP-1R (orange) residues that interact with GLP-1 (green) or waters (red balls) are displayed in stick format; middle: GLP-1R (pale green) and PF 06882961 (purple); bottom: GLP-1R (pale blue) and CHU-128 (yellow). Two views are depicted for clarity: left, TM bundle viewed from the upper portion of TM6/TM7 with TM6-ECL3-TM7 removed, and right, TM bundle viewed from the upper portion of TM4/TM5 with TM4-ECL2-TM5 removed. Dashed lines depict hydrogen bonds as determined using UCSF Chimera. See also Figures 2 and S3–S5.

GLP-1 first requires molecular accuracy in understanding of how GLP-1 itself binds to the receptor, including the involvement of structural waters in mediating interactions. Our GLP-1-bound complex provides a 2.0 Å increase in resolution over the previously published active GLP-1:rabbit GLP-1R:G_s complex (Zhang et al., 2017). This structure is generally similar to ours, particularly toward the intracellular face, but there are substantial differences in the models for the peptide binding site, ECLs, and the top of TM1, consistent with difficulty in modeling these regions where there was limited resolution in the 4.1-Å map (Figure S4A). As such, details of GLP-1 binding that are well supported by density in our cryo-EM map (Figures S3 and S4) are described below.

$H7^{GLP-1}$ and $A8^{GLP-1}$ are crucial for GLP-1 function, as truncation of these two residues, forming the GLP-1(9-36) NH_2 analog, lowers affinity and efficacy (Montrose-Rafizadeh et al., 1997). These two residues reside deep within the TM cavity and form extensive interactions with the GLP-1R (Figure 3). $H7^{GLP-1}$ interacts with TM3 and TM5 through hydrogen bond and hydrophobic interactions with $Q234^{3.37}$, $V237^{3.40}$, $W306^{5.36}$, $R310^{5.40}$, and $I313^{5.43}$, whereas $A8^{GLP-1}$ forms hydrophobic interactions with $E387^{7.42}$ and $L388^{7.43}$ in TM7 (Figures 3 and S4). In addition, $H7^{GLP-1}$ is stabilized through a hydrogen bond of its backbone with the side chain of $T11^{GLP-1}$ and through a structural water that interacts with the backbone of TM5. $E9^{GLP-1}$ also resides at the base of the binding pocket participating in van der Waals interactions with $R190^{2.60}$ and $L388^{7.43}$ as well as hydrogen bond interactions with $R190^{2.60}$ and $Y152^{1.47}$, both crucial residues for GLP-1R activation and signaling (Lei et al., 2018; Wootten et al., 2016a, 2016b). In addition, $A8^{GLP-1}$ (backbone) and $E9^{GLP-1}$ engage in an extensive hydrogen bond network with structural waters and polar residues at the base of the



(legend on next page)

peptide binding pocket (R190$^{2.60}$, Y152$^{1.47}$, and Y241$^{3.44}$), as well as with the backbone of TM7 (Figures 3 and S4). This likely stabilizes the peptide in the base of the pocket and helps maintain an active receptor conformation, as alanine mutation of these receptor side chains reduced both affinity and efficacy of GLP-1 and other peptide ligands (Figure S5; Lei et al., 2018; Wootten et al., 2016a, 2016b). GLP-1 also forms extensive interactions with ECLs 2 and 3 where S14$^{GLP\text{-}1}$ and S17$^{GLP\text{-}1}$ interact via both hydrogen bonding and van der Waals interactions with T298^{ECL2}, R299^{ECL2}, and N300^{ECL2}. T11$^{GLP\text{-}1}$ and D15$^{GLP\text{-}1}$ form polar and hydrophobic interactions with D372^{ECL3}, R380$^{ECL3/7.35}$, and L384$^{7.39}$ (Figures 3 and S4). These positions within ECL2 and 3 are crucial for peptide affinity and receptor activation in numerous class B1 GPCRs (Coin et al., 2013; Koole et al., 2012; Woolley et al., 2017). Other notable interactions include multiple hydrophobic contacts of the peptide with residues in TM1 (L141$^{1.36}$, L144$^{1.39}$, and Y148$^{1.43}$), hydrogen bonding between K197$^{2.67}$ and T13$^{GLP\text{-}1}$, Y205$^{2.75}$ with both S17$^{GLP\text{-}1}$ and E21$^{GLP\text{-}1}$ and interactions of E21$^{GLP\text{-}1}$ with the far N terminus of the receptor. The importance of these key interactions in GLP-1 function are supported by extensive alanine mutagenesis studies (Figure S5; Coopman et al., 2010; Dods and Donnelly, 2015; Graaf et al., 2016; Lei et al., 2018; Wootten et al., 2013b, 2016c; Yang et al., 2016).

PF 06882961, but Not CHU-128, Mimics Key Features of the GLP-1 Binding Linked to Receptor Activation and Biased Agonism

Relative to GLP-1, both small-molecule agonists bind superficially to the transmembrane domain of the receptor but adopt very distinct poses where they engage in different receptor interactions (detailed in Figures 3 and S4 and Table S2) and induce distinct conformations within the receptor extracellular domain (Figure 4). Close examination of these structures, and comparison to the GLP-1-bound structure, provides mechanistic insights into how PF 06882961, but not CHU-128, can closely mimic the pharmacology of GLP-1.

CHU-128 exhibited a planar pose (parallel to the membrane) in the receptor TM bundle, with overlap with GLP-1 limited to residues F12$^{GLP\text{-}1}$-L16$^{GLP\text{-}1}$. The fluoro methyl indazole and dimethyl morpholine arms extend beyond the pocket occupied by peptide and PF 06882961 (Figure 4). Despite this, the majority of side chains that interact with CHU-128 also interact with GLP-1 (Figure 3; Table S2), including P137$^{1.32}$, E138$^{1.33}$, L141$^{1.36}$, L144$^{1.39}$, Y145$^{1.40}$, Y148$^{1.43}$, K197$^{2.67}$, L201$^{2.71}$, Y205$^{2.75}$, F230$^{3.33}$, M233$^{3.36}$, T298^{ECL2}, N300^{ECL2}, and L388$^{7.43}$, albeit the nature of these interactions were distinct and, unlike GLP-1, CHU-128 did not form strong interactions with TM7. In contrast, PF 06882961 docked in a buried pocket within the receptor, adopting an elongated pose principally located within the bound GLP-1 molecular envelope, where it overlaps the location of GLP-1 residues G10$^{GLP\text{-}1}$-E21$^{GLP\text{-}1}$ in the GLP-1-bound structure (Figure 4). This leads to a subset of ligand interactions with shared receptor residues, including S31^{ECD}, L32^{ECD}, L141$^{1.36}$, K197$^{2.67}$, L201$^{2.71}$, Q221^{ECL1}, F230$^{3.33}$, M233$^{3.36}$, T298^{ECL2}, R380$^{ECL3/7.35}$, and L384$^{7.39}$ (Figures 3 and S4; Table S2). Nonetheless, even within this region, GLP-1 has greater bulk that confers distinct interactions, including in engagement with ECL2, ECL1, TM7, and ECL3 (Figure 4). These structures raise the question: how does PF 06882961 closely mimic the pharmacology of GLP-1, although CHU-128 only productively engenders a subset of these responses, specifically cAMP production?

PF06882961 Stabilizes an Extensive Water-Mediated Hydrogen Bond Network within the GLP-1R TM Binding Cavity

As noted above, potent canonical signaling by GLP-1 requires the far N-terminal dipeptide H7$^{GLP\text{-}1}$-A8$^{GLP\text{-}1}$ that resides deep within the receptor TM binding cavity and makes extensive interactions with receptor residues known to contribute to receptor activation. Remarkably, the PF 06882961 binding cavity extends below the ligand deep into the receptor core, and there is overlap of this deep pocket with the binding site of the GLP-1 N-terminal H7$^{GLP\text{-}1}$-E9$^{GLP\text{-}1}$ and the waters below the peptide that facilitate interaction with the central polar network of the receptor (Figures 3 and 4E). Moreover, the structural differences at the extracellular face of the receptor between the PF 06882961 and GLP-1-bound structures (discussed in detail below) ensure that this deep pocket is not exposed to bulk solvent in the PF-06882961-bound structure (Video S1). Consequently, there is a well-resolved structural water network that extends below the PF 06882961 binding site (Figure 3) that directly overlaps the location of H7$^{GLP\text{-}1}$-E9$^{GLP\text{-}1}$ and stabilizes a similar conformational arrangement of the GLP-1R residues deep within the binding pocket, including those that interact directly, or via water, with GLP-1 (Figure 4E). Although the binding cavity also extends below the bound CHU-128 ligand, CHU-128 induces a larger

Figure 4. Comparisons of the GLP-1R Conformations and Binding Pockets Stabilized by GLP-1 and Non-peptidic Agonists

(A–C) Superimposition of the receptor from the GLP-1R:G_s complex structures with GLP-1 (orange), PF 06882961 (pale green), CHU-128 (pale blue), and TT-OAD2 bound (6ORV, cornflower blue).

(A) Overlay of full-length receptors; (B) close up of the ECD showing the distinct location of the ECD alpha helix in each structure; (C) close up of the GLP-1R TM bundle viewed from the extracellular side reveals differences in the conformations of TM1, TM2/ECL1, and the TM6-ECL3-TM7 backbones when stabilized by different agonists.

(D) Left: Superimposition of the TMD from the PF 06882961- (pale green) and the CHU-128-bound (pale blue) receptors reveals partial but limited overlap in the ligand binding pose for PF 06882961 (purple) and CHU-128 (yellow); right: side view of the overlaid upper portions of the TM bundles for GLP-1R bound to GLP-1 (receptor orange; GLP-1 green), PF 06882961 (receptor pale green; ligand purple), and CHU-128 (receptor pale blue; ligand yellow) reveals both non-peptide ligands overlap the binding of GLP-1 but stabilize different conformations in the TM bundle.

(E) PF 06882961 (purple) has a larger degree of overlap with GLP-1 (green) compared with CHU-128 (yellow). The water network below the bound PF 06882961 (purple spheres) overlaps the N terminus of GLP-1 and forms an extensive hydrogen bond network with polar residues deep in the TM binding cavity. The CHU-128 complex also contains structural waters (yellow spheres) that interact with polar residues deep in the GLP-1R TM cavity or with ligand.

See also Figures 5, 6, S5, and S6 and Video S1.

binding cavity with a less-extensive hydrogen bond network with stabilized waters (Figures 3 and 4E; Video S1).

Distinct GLP-1R Conformations Are Stabilized by Agonists with Differing Pharmacology

Extensive GLP-1R alanine mutagenesis studies have revealed that ECL2 and TM6/ECL3/TM7 are critical for peptide-mediated G_s-dependent cAMP signaling, although TM1/TM7/ECL3 are critical for pERK1/2 that is a composite of β-arrestin-dependent and non-canonical G-protein-mediated signaling (Lei et al., 2018; Wootten et al., 2016c). Together with mutagenesis of residues deep within the receptor binding domain (Figure S5; Graaf et al., 2016; Wootten et al., 2013b; Yang et al., 2016), this provides a framework for interpreting key features of the GLP-1R:G_s structures to understand characteristics linked to the common signaling of GLP-1 and PF 06882961 and to understand the distinct functional profile of CHU-128, as well as TT-OAD2 (through the recently reported GLP-1R structure [Zhao et al., 2020] with this compound).

The recently published inactive structure of the full-length GLP-1R (PDB: 6LN2; Wu et al., 2020), which likely closely correlates to the apo state, allows greater understanding of the transitions that occur within the GLP-1R as it moves from the inactive conformation to different active states (Video S2). In the inactive state, the GLP-1R ECD is stabilized in a "closed conformation" with the ECD-peptide-binding groove oriented toward the ECLs through interactions of the ECD N-terminal helix with ECL3 and the top of TM7 and the ECD loop region between S116 and E127 with ECL1. ECL1 itself provides an additional cap to the receptor core as well as forming interactions with ECL2 that likely constrain the conformation of this latter domain. The importance of the interaction between the far ECD and ECL3 for maintaining receptor quiescence is supported by mutagenesis and biochemical studies with GLP-1R and the related glucagon receptor (GCGR) that reveal loss of this interaction can reduce the barrier to activation or engender constitutive activity in the case of the GCGR (Wu et al., 2020; Yang et al., 2015; Yin et al., 2016; Zhao et al., 2016). Nonetheless, molecular dynamics simulations indicate that the ECD is dynamic and can transiently, partially or fully, disengage from the receptor core, and this is likely key to both peptide and small-molecule binding (Wu et al., 2020; Yang et al., 2015; Zhang et al., 2019; Zhao et al., 2020).

During receptor activation, there is release of the interactions between the ECD and receptor core and an upward and clockwise rotation of the ECD that is greatest upon accommodation of GLP-1 (Figure 5A; Video S2). In the small-molecule-bound structures, this ECD translation is more limited, and the far N-terminal helix caps the binding site for both PF 06882961 and CHU-128. In these conformations, the ECD maintains interactions with the ECLs, particularly ECL1 and ECL2, in addition to the extreme ECD forming interactions with TM7 (PF 06882961) or TM1 (CHU-128; Figure 3). An outward movement of ECL1 that is enabled by the reorganization of the TMD:ECD interactions also occurs (Figure 5). For the non-peptidic agonists, the extent of ECL1 translation is dictated by the ligand-binding pose and the interactions of the ECD with TMs 1 or 7. The dimethyl morpholine arm of CHU-128 extends toward ECL1, leading to a greater outward translocation of this domain relative to that seen in the PF-06882961-bound structure (Figures 3 and 4D). Nonetheless, in both small-molecule-bound structures, the conformation of ECL1 is similar to that observed in the inactive structure, whereas it undergoes reorganization to accommodate GLP-1 binding (Figure 5; Video S2).

Interestingly, although the overall conformation of ECL2, one of the most critical regions involved in G_s-mediated signaling, is similar in the active GLP-1 and non-peptide-bound structures, there are distinctions in how this is maintained (Figure 3). In the GLP-1-bound structure, direct interaction between side chains located above and below ECL2 constrain the conformation of this domain. In contrast, there is limited direct interaction between PF 06882961 and ECL2; however, the conformation is instead constrained by water-mediated interactions between ECL1, the ECD, particularly $W33^{ECD}$, and ECL2. Although the CHU-128 compound has a very distinct docking pose, ECL2 is stabilized by ligand interactions, as well as similar interactions between the ECD and ECL2 (Figure 3), and these interactions likely play a similar role in supporting G_s-mediated signaling. In the previously reported TT-OAD2 structure, ECL2 also adopts a similar "active" conformation that would support G_s coupling and activation.

The greatest diversity across the GLP-1-, PF-06882961-, and CHU-128-bound structures is in the location and conformation of TM6/ECL3/TM7 and TM1 (Figures 4 and 5). Although speculative, it is likely that disengagement of the interaction of the ECD N-terminal helix with ECL3 present in the "apo-like" structure leads to increased mobility and outward movement of this domain, as reflected in inactive structures of class B1 GPCRs that lack the ECD and in the TT-OAD2-bound structure where the ligand does not directly interact with this receptor domain. As such, ligands that form direct interactions with the top of TM7/ECL3 lead to an overall contraction of the binding pocket relative to those that do not. The key structural differences in the positions and/or conformations of TM7, ECL1, and the ECD observed between the GLP-1-, PF-06882961-, and CHU-128-bound structures (Figures 4 and 5) contribute to both the ability of PF 06882961 to mimic the action of GLP-1 and the divergence in signaling and regulation seen with CHU-128.

PF 06882961 Stabilizes TM7 in a Conformation that Enables It to Mimic Key Interaction Networks that Are Stabilized by GLP-1 Binding

In the PF-06882961-bound structure, TM7 is translated dramatically inward, overlapping with the molecular envelope of GLP-1 binding. This both stabilizes compound binding and seals the deeper pocket from access to bulk water (Figures 3 and 4; Video S1). Intriguingly, despite the large contraction of the upper part of TM7 induced by the binding of PF 06882961, this region along with the structural waters enables interaction networks within the receptor that can mimic the GLP-1-induced active GLP-1R conformation (Figures 3 and 6). In the GLP-1-bound structure, the more outward conformation of ECL3 and TM7 is stabilized by an extensive polar network that extends from $D15^{GLP-1}$ that forms a salt bridge with $R380^{ECL3/7.35}$. From this ligand interaction, $R380^{ECL3/7.35}$, $D372^{ECL3}$, $W306^{5.36}$, $R310^{5.40}$, and $E373^{ECL3}$ form consecutive interactions via a salt bridge, hydrogen bond, cation-π stacking, and salt bridge, respectively,

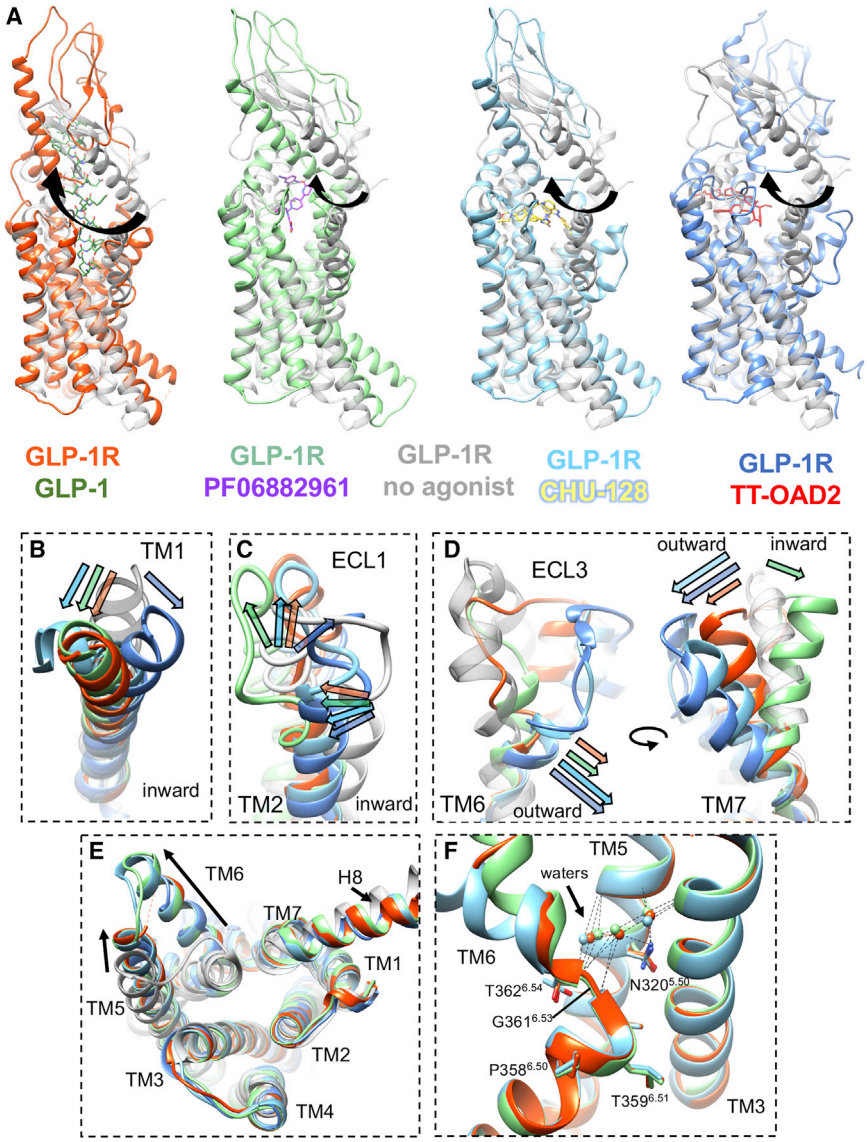


Figure 5. GLP-1R Conformational Transitions from the Inactive Conformation to Different Active States

All panels: superimposition of the inactive, ligand-free GLP-1R (PDB: 6LN2, pale gray) with the GLP-1R bound to GLP-1 (orange), PF 06882961 (pale green), CHU-128 (pale blue), and TT-OAD2 (6ORV, cornflower blue). Arrows depict direction of movement when comparing inactive (gray) to agonist-bound states.

(A) The ECD undergoes a large conformational transition upon binding of all agonists, but the degree of movement differs depending on the bound ligand.

(B and C) TM1 and TM2 move inward toward the center of the TM bundle, and ECL1 moves upward in the presence of all agonists; however, the exact nature of these movement differs.

(D) The top of TM6 unwinds in the presence of all agonists, and CHU-128 and TT-OAD2 promote a significant outward movement. Large outward movements within ECL3 and TM7 are also observed in CHU-128- and TT-OAD2-bound GLP-1Rs and to a lesser extent GLP-1; however, PF 06882961 promotes a significant inward movement of TM7 and ECL3.

(E) At the intracellular face, all agonists promote the same receptor conformation that involves a large outward movement of TM6 and a smaller outward movement of TM5. There is also a small downward movement of H8 away from the detergent micelle.

(F) The TM6 kink around the conserved $P^{6.50}$-X-X-$G^{6.53}$ motif is stabilized by a similar network of residues and structural waters in all active state structures (spheres depict waters colored relative to their bound ligand, with hydrogen bonds depicted by dotted lines).

See also Figures 4, S5, and S6 and Video S2.

where all but $E373^{ECL3}$ also form interactions with the GLP-1 peptide (Figure 6A). This latter residue, however, is important for the differential engagement of GLP-1 and the biased agonist peptide, ExP5, demonstrating importance of the ECL3 conformation and signaling efficacy (Zhao et al., 2020). Furthermore, the ECL3 interaction network, mediated via $K383^{7.38}$, interconnects with the water network that stabilizes the N terminus of GLP-1 ($H7^{GLP-1}$-$E9^{GLP-1}$) and the central polar core of the receptor (Figures 3 and 6A). Alanine mutation of $R310^{5.40}$, $K383^{7.38}$, $D372^{ECL3}$, and $E373^{ECL3}$ reduce the potency of GLP-1 by over 1,000-fold in multiple signaling pathways (cAMP, calcium, and pERK1/2), although these mutations have a much smaller effect on affinity (Coopman et al., 2010; Wootten et al., 2016b, 2016c). Together, the structural and mutagenesis studies support the strong influence of this polar interaction network in GLP-1-mediated signal transduction and stabilization of the overall active conformation.

Together with interactions directly with the bound PF 06882961 scaffold, the inwardly translated TM7 conformation is able to mimic the polar interaction network observed in the GLP-1-bound structure (Figures 3 and 6A). Here, the carboxylate of PF 06882961 mimics $D15^{GLP-1}$ of GLP-1 as they both share an ionic interaction with $R380^{ECL3/7.35}$. The inward movement of TM7 is followed by TM6 enabling $E373^{ECL3}$ to form the same salt bridge with $R310^{5.40}$ that was formed in the GLP-1 complex (Figure 6A). These changes allow other shared polar residues to move and partake in the very extensive water-supported polar network that links the bound ligand to the central polar network (Figures 3 and 4). This includes $R299^{ECL2}$ that moves inward to form a salt bridge with $E373^{ECL3}$ and cation-π stacking with $W306^{5.36}$, thereby aiding the stabilization of TM6 and TM7 in the contracted conformation of the vestibule (Figure 6A). In support of this, alanine mutation of $R380^{ECL3/7.35}$ or $E373^{ECL3}$ had a large impact on cAMP production mediated by both GLP-1 and PF 06882961 (Figure S5). Furthermore, the TM7 contraction induced by PF 06882961 moves $F385^{7.40}$ into a hydrophobic groove between TM1 and

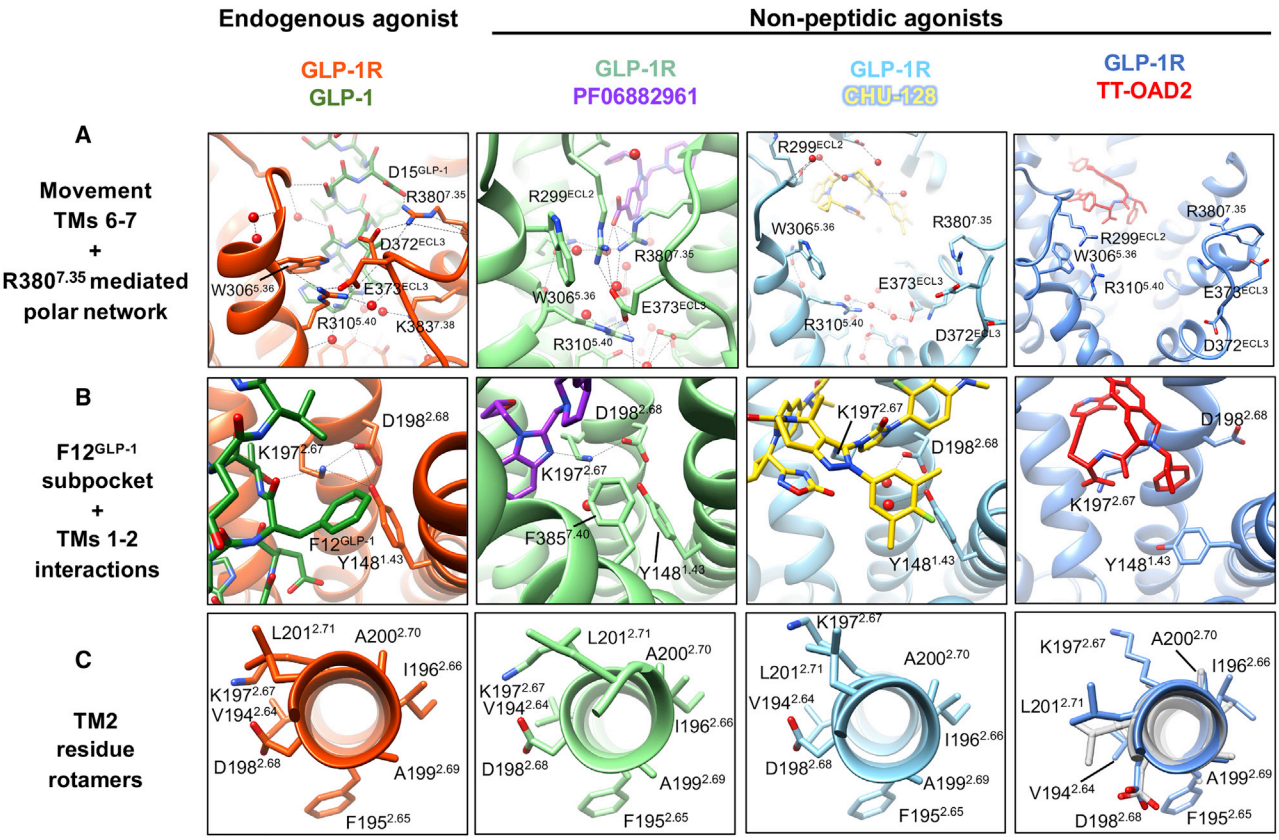


Figure 6. Distinct Interaction Networks Provide a Structural Rationale for How PF 06882961, but Not CHU-128, Can Mimic the Signaling of GLP-1

Top: the conformation of TM6/ECL3/TM7 in the GLP-1- (far left) and PF 06882961-bound (middle left), but not the CHU-128- and TT-OAD2-bound (right panels) GLP-1Rs, enables the formation of an extensive polar interaction network. This is mediated by interactions of $R380^{7.35}$ with $D15^{GLP-1}$ of the peptide or the carboxylic acid of PF 06882961. In the PF 06882961, this network stabilizes the water-mediated network below the bound ligand that overlaps the peptide binding site. Middle: the TM7 contraction induced by PF 06882961 moves $F385^{7.40}$ into a hydrophobic groove between TM1 and TM7 (middle left), occupying the same pocket filled by $F12^{GLP-1}$ in the peptide-bound structure (far left), thereby stabilizing common overall rearrangements within side chains of key residues located in TMs 1 and 2. The fluoro dimethyl phenyl moiety of CHU-128 plays a similar role in the CHU-128-bound structure (middle right), albeit different interactions of the ligand with $K197^{2.67}$ stabilize a different conformation of this side chain. TT-OAD2 cannot fill this pocket and does not form this polar network (far right). Bottom: in the absence of the polar network, residues in TM2 adopt a different rotameric arrangement in the TT-OAD2-bound receptor, which is close to that observed in the inactive receptor (gray) when compared with GLP-1, PF 06882961, and CHU-128. See also Figures 4, S3, S5, and S6.

TM7 to occupy the pocket filled by $F12^{GLP-1}$ in the peptide-bound structure, thereby stabilizing common overall rearrangements of side chains of key residues located in TMs 1 and 2 (Figures 3 and 6B). Both $F385^{7.40}$ in the PF 06882961 structure and $F12^{GLP-1}$ of GLP-1 show T-shaped π-stacking interactions with $Y148^{1.43}$, which undergoes a rotameric change relative to inactive structures, leading to a hydrogen bond with $D198^{2.68}$. $D198^{2.68}$, in turn, forms a salt bridge with $K197^{2.67}$, and these side chains influence the rotamers of surrounding residues (Figure 6C), which either engage with peptide or with the water network below the PF 06882961 binding site (Figure 3). The importance of $F385^{7.40}$ in PF 06882961 activity was confirmed by alanine mutagenesis, where this mutation reduced PF 06882961 potency by 10-fold with no influence on GLP-1 cAMP production (Figure S5). In addition, $Y148^{1.43}$, $K197^{2.67}$, and $D198^{2.68}$ are also important for ligand interactions and activation as shown by large potency changes in mutational studies (Figure S5; Coopman et al., 2010; Lei et al., 2018; López de Maturana and Donnelly, 2002). The similarities in the interaction networks formed within the GLP-1- and PF-06882961-bound receptor result in similar electrostatic and hydrophobic properties of their binding pockets, despite differences in their overall shape and volume (Figure S6; Video S1).

The CHU-128-Bound GLP-1R, but Not TT-OAD2, Is Able to Partially Mimic the Binding Pocket Hydrogen Bond Networks Induced by GLP-1 and PF 06882961

CHU-128 is a potent agonist of the GLP-1R for cAMP production, and this contrasts with TT-OAD2 that is a low efficacy agonist for this pathway (Zhao et al., 2020; Figure 7). The fluoro methyl indazole arm of CHU-128 extends toward TM1 and TM2, with the fluoro dimethyl phenyl moiety π stacking with $Y148^{1.43}$ in a similar manner to $F12^{GLP-1}$, and $F385^{7.40}$ in the PF 06882961 structure, and this supports similar polar interactions with and

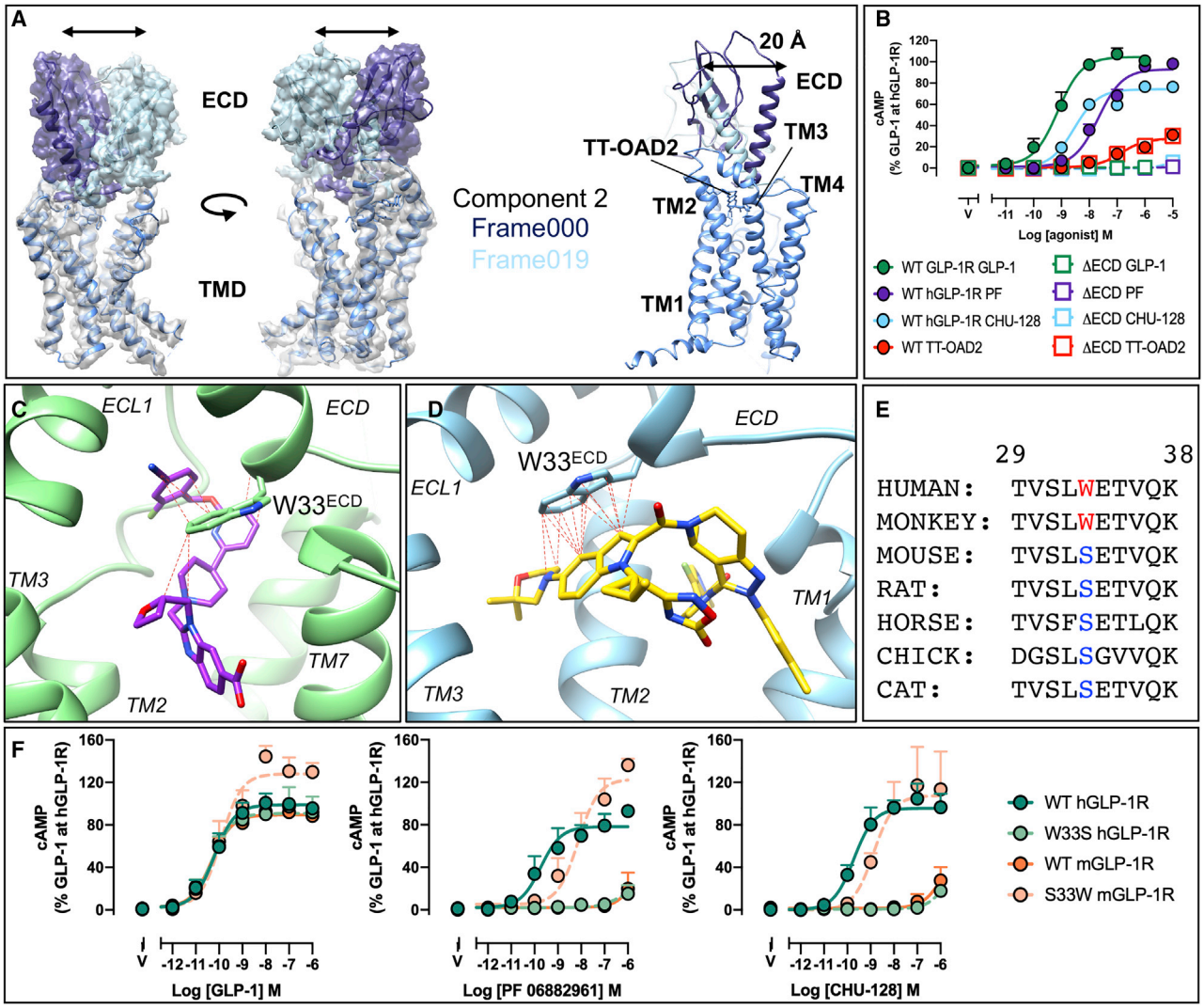


Figure 7. The GLP-1R ECD Differentially Influences the Pharmacological Profiles Observed for GLP-1R Agonists

(A) 3D variability analysis using cryoSPARC: left orthogonal views of the cryo-EM maps from frames 000 and 019 from component 2 of the variability analysis show extensive mobility in the ECD relative to the TM bundle of the TT-OAD2-bound GLP-1R that was not observed with PF 06882961 or CHU-128 (see Video S4). Measurement of the distance between the Cα of E52 (of ECD models rigid body fit to the two extreme maps) revealed a 20-Å movement of the ECD.

(B) cAMP production concentration-response curves for GLP-1 (green), PF 06882961 (purple), CHU-128 (pale blue), and TT-OAD2 (red) at the full-length GLP-1R (closed circles) and the "headless" GLP-1R that lacks the ECD (open squares), show the ECD is not required for the partial agonism exhibited by TT-OAD2 but is essential for the activity of GLP-1, PF 06882961, and CHU-128.

(C and D) PF 06882961 (C) and CHU-128 (D) form extensive contacts with $W33^{ECD}$ within the ECD of the GLP-1R.

(E) $W33^{ECD}$ within the ECD is conserved in human and primate GLP-1R, but not rodents and other species.

(F) cAMP concentration response curves for GLP-1 (middle left), PF 06882961 (middle), and CHU-128 (middle right) at the human (h)GLP-1R and mouse (m)GLP-1R reveal species selectivity of PF 06882961 and CHU-128 with no response at the mouse GLP-1R. Swapping the residue at position 33 between hGLP-1R and mGLP-1R (W33S hGLP-1R and S33W mGLP-1R) switches the species selectivity. Pharmacological data are presented as the mean + SEM of at least 3 independent experiments performed in triplicate or duplicate.

See also Figure S5 and Videos S3 and S4.

within TM2. By comparison, although there is considerable overlap with the pose, TT-OAD2 lacks an aromatic extension equivalent to the CHU-128 fluoro dimethyl phenyl group, resulting in $Y148^{1.43}$ and $D198^{2.68}$ adopting distinct rotameric positions that cannot form an equivalent polar network to constrain the TM1-TM2 interface (Figure 6B). As these residues play a critical role in ligand potency of the other small-molecule agonists, these observed conformational differences may contribute to the lower efficacy of TT-OAD2.

In contrast to the GLP-1- and PF-06882961-bound GLP-1R structures, there is only very limited interaction of CHU-128 with TM7. This leads to an outwardly splayed conformation of TM7/ECL3/TM6, similar to that observed in the TT-OAD2-bound GLP-1R (Figures 4A and 4C). This position of TM6/ECL3/TM7

exposes the deep binding pocket below CHU-128 to bulk water (Figures 3 and 6A). Density in the cryo-EM map corresponding to waters was present deep in the CHU-128 binding pocket, where they stabilize the rotameric positions of the deep central polar network of the receptor (Figure 4E). However, unlike the PF-06882961-bound structure, these did not form a continuous network extending to the ligand (Figures 3 and 4E), albeit additional weak density that could not be confidently assigned to waters was also present in the cryo-EM map. Despite the overall higher resolution of the cryo-EM map within the CHU-128 ligand binding site (relative to PF 06882961), the less-well-resolved water density is consistent with greater mobility of waters in the larger binding cavity and the lack of the extensive polar interactions extending from TM7 that were observed in the PF-06882961-bound structure (Figure 6A). These differences also lead to distinctions in the hydrophobicity and electrostatic potential of the deeper binding cavity of the CHU-128 binding pocket (Figure S6A). The absence of the hydrogen bond network within TM7-TM6-TM5 may account for the lower efficacy for the cAMP pathway observed with CHU-128, relative to PF 06882961. Indeed, alanine mutation of $E373^{ECL3}$, one of the key mediators of this network in the GLP-1- and PF-06882961-bound structures, had limited influence on CHU-128 when compared to PF-06882961- and GLP-1-mediated cAMP production (Figure S5).

The TM6-ECL3-TM7/TM1 Domain Is Associated with GLP-1R-Biased Agonism

The open conformation of TM6-ECL3-TM7 may also account for the biased agonism observed with CHU-128 and TT-OAD2. This is consistent with mutagenesis studies demonstrating the importance of TM1 and TM6-ECL3-TM7 in biased agonism of peptide agonists (Lei et al., 2018; Wootten et al., 2016c) and is further supported by the structure of the GLP-1R bound to ExP5 (Liang et al., 2018b; Zhao et al., 2020). In this latter structure, the TM6-ECL3-TM7 region resides in an intermediate position to that induced by GLP-1 and by CHU-128 and TT-OAD2 (Figure S6C). ExP5 is a G-protein-biased peptide relative to GLP-1 with greater coupling to cAMP and calcium mobilization relative to β-arrestins and receptor internalization, where it is only a partial agonist (Zhang et al., 2015). Thus, the relative positioning of the TM6-ECL3-TM7 domain to the rest of the bundle is correlated to the ability to couple to regulatory pathways, with a more open conformation resulting in reduced responses. This provides a potential structural basis for the differences in the signaling and regulatory profile of CHU-128 (and TT-OAD2 and ExP5) relative to GLP-1 and PF 06882961.

GLP-1R Agonists All Promote a Similar GLP-1R Conformation below the Ligand Binding Cavity

Despite differences in the extracellular conformation induced by GLP-1 and non-peptide agonists, these all converge to similar conformation at the base of the binding pocket, with a similar reorganization of polar residues that enables the formation and stabilization of the TM6 kink, a hallmark of the class B1 GPCR active state, and includes structural waters (Figure 5). Although the resolution of the published TT-OAD2-bound structure did not allow for the visualization of waters (Zhao et al., 2020), the almost identical side-chain rotamers suggest the same stabilization networks exist. All agonist-bound structures adopt an equivalent conformation within the GLP-1R intracellular face, with the most predominant feature being a large outward movement of TM6 relative to the inactive receptor, consistent with previous observations for class B1 GPCR active-state structures (Figure 5; Liang et al., 2017, Liang et al., 2018a, Liang et al., 2018b, Liang et al., 2020a, Liang et al., 2020b; Ma et al., 2020; Qiao et al., 2020; Zhang et al., 2017; Zhao et al., 2019, 2020). These common interaction networks in the center and base of the receptor, regardless of the agonist, are likely to be allosterically influenced by the presence of the bound G protein. The interactions formed between the G_s protein and the receptor were equivalent between the GLP-1-, PF-06882961-, and CHU-128-bound complexes and consistent with previously published GLP-1R:G_s complex structures (Liang et al., 2018b; Zhang et al., 2017; Zhao et al., 2020), albeit additional interactions (including water mediated) were identified in the current study due to the higher resolution (Figure S7).

Collectively, these data reveal structural similarities and differences associated with peptide and non-peptide agonist binding and provide mechanistic insight into how PF 06882961 mimics key conformational features of the GLP-1-bound receptor to engender a similar signaling profile, while also providing confirmatory evidence for the role of the TM1/TM7/ECL3/TM6 domain in engendering biased agonism.

Receptor and G Protein Dynamics Contribute to Agonist-Dependent G Protein Coupling and Activation

3D variability analysis in cryoSPARC (Punjani et al., 2017) was used to extract the 3 major principal components that contributed to conformational variance for each of the structures. This revealed common, coordinated twisting and rocking motions of the receptor, relative to the G protein, for all 3 structures (Video S3). Notably, equivalent analysis of the previously solved TT-OAD2 structural data suggested that this complex has a more stable receptor-G protein interface with lower relative motion (Video S3). In contrast to the more efficacious agonists for G_s activation, the α-helical domain (AHD) of $G\alpha_s$ in the TT-OAD2 complex occupies a single, open orientation, albeit that this was poorly resolved, indicative of substantive motion within this location (Video S4). For the other complexes, the AHD undergoes translational motion between open and more closed orientations, relative to the $G\alpha_s$ Ras-like domain (Video S4). Collectively, the predominant open conformation of $G\alpha_s$ and more limited dynamics of the receptor-G protein interface in the TT-OAD2 structure are consistent with the slow G_s activation kinetics and rate of cAMP production previously observed for this ligand, though additional experiments are required to better understand the unique pharmacological behavior of TT-OAD2.

Due to the conformational variance of the $G\alpha_s$ AHD in the GLP-1-, PF-06882961-, and CHU-128-bound complexes, we performed additional G protein focused refinements that enabled the identification of five distinct classes with differences in the AHD position for each complex (Figure S7; Video S4). Three classes were common to all three datasets (labeled up [most open], middle, and down [most closed]). The number of particles present in each class differed for the three agonists, suggestive of

different mobilities and conformational ensembles stabilized by each ligand. Although not the most populated class, the "down" AHD position was the best resolved and the only class that allowed confident modeling into the map (Figure S7). Although the distinct sampling of AHD position across the structures would also be consistent with each of the agonists having discrete efficacy, it is important to note that the biochemical system used for complex formation is artificial (detergent micelle, dominant-negative $G\alpha_s$, and Nb35) and that extrapolation of observed behavior to the normal cellular membrane environment needs to done with a high degree of caution.

In line with the lower resolution of ECL1 and the ECD in consensus maps of the GLP-1-bound receptor, these domains exhibited greater motion relative to the rest of the bundle compared to the equivalent domains in the PF-06882961- and CHU-128-bound receptors (Figure S2; Videos S3 and S4). Modeling the GLP-1R backbone into the two extreme frames of the analysis revealed a coordinated 7-Å movement in these regions (Video S2) and a 4.5-Å movement of the location of the C terminus of the GLP-1 peptide, although the N-terminal region that binds deep within the TM bundle and is responsible for activation is relatively stable. Interestingly, in the TT-OAD2-bound GLP-1R, ECL1 is relatively stable, although the ECD is extremely mobile with more substantial motions than those induced by GLP-1, transitioning 20 Å between the two extreme frames of the variance analysis when measured from the Cα of $E52^{ECD}$ in models built into the maps (Figure 7A; Video S3). This mobility is also reflected in the low resolution of the GLP-1R ECD in the consensus map TT-OAD2-GLP-1R-G_s complex (Zhao et al., 2020). This substantial flexibility is likely related to the lack of stabilizing interactions formed between the ligand and the ECD and a lack of interaction of the ECD with ECLs1/2 and TMs1/7 that were present in the PF 06882961 and CHU-128 structures. Previous studies revealed the GLP-1R ECD contributes to agonist-induced cAMP signaling (Yin et al., 2016). In line with this, we identified the ECD is crucial for the function of GLP-1, PF 06882961, and CHU-128 but is not required for the cAMP signaling mediated by TT-OAD2 (Figure 7B). The ability of PF 06882961 and CHU-128 to engender a stable ECD conformation more akin to that of peptide agonists, when compared to TT-OAD2, may therefore also contribute to their higher efficacy in cAMP production.

PF 06882961 and CHU-128 Have Species Selectivity that Can Be Explained by Their Interaction with the ECD

PF 06882961 and CHU-128 form extensive hydrophobic interactions with $W33^{ECD}$ within the N terminus of the receptor ECD (Figures 7C and 7D; Table S2). Mutation of W33 to alanine completely abolished cAMP production by both agonists, with no effect on GLP-1-mediated signaling (Figure S5). Tryptophan at position 33 is only conserved in human and primate GLP-1R, but not in other mammalian species, where it is a serine (Figure 7E). In cell lines expressing rodent receptors, CHU-128 and PF 06882961 were unable to promote cAMP production, revealing that these compounds have species selectivity (Figure 7F). Mutation of $W33^{ECD}$ to serine in human GLP-1R had a similar effect to the alanine mutation, abolishing cAMP signaling; however, a serine-to-tryptophan mutation in rodent receptors was sufficient to recover compound activity (Figure 7F), confirming the essential role of $W33^{ECD}$ in this species selectivity. This has significant implications for preclinical development of GLP-1R small-molecule agonists, as these are often assessed in rodent models of type 2 diabetes and obesity.

Concluding Remarks

Diabetes and obesity are major health burdens, and the GLP-1R is a validated target to regulate metabolism for treatment of these diseases (Brown et al., 2018; Nauck and Meier, 2019). Despite major interest, development of orally delivered non-peptide GLP-1R drugs has proven profoundly difficult, and only three compound series have progressed into clinical development. We have identified unique activation profiles for non-peptide ligands from these series and differences in how they bind to and activate the GLP-1R. The cryo-EM structures provide a deeper understanding of the active-state conformations that the GLP-1R is capable of adopting and reveal new druggable pockets for the development of small-molecule agonists. The near-atomic level detail revealed within the binding pockets induced by GLP-1, PF 06882961, and CHU-128 provide critical information that can be utilized in structure-guided-activity programs to progress development of the next generation of GLP-1R non-peptide agonists. Moreover, PF 06882961 is able to better mimic the *in vitro* pharmacological actions of GLP-1 than some closely related peptides, making this a particularly attractive lead structure, and this activity can be rationalized from the new high-resolution structures. This could potentially provide a template for development of higher affinity/efficacy compounds than those currently available. Furthermore, the spectrum of non-peptide agonist binding sites identified herein for the GLP-1R may extend to other class B1 GPCRs that are of high therapeutic potential. These receptors are likely capable of adopting similar conformations if targeted by appropriate small-molecule scaffolds, albeit the specific nature and properties within these pockets may differ between receptors due to their differing sequences. Our work opens up new avenues to pursue small-molecule agonists identification and development.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.molcel.2020.09.020.

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AUTHOR CONTRIBUTIONS

D.W. and P.M.S. conceived and formulated the research. X.Z. expressed and purified complexes; R.D., M.J.B., and H.V. performed sample vitrification and cryo-EM imaging; X.Z., M.J.B., and R.D. processed the EM data; X.Z. and M.J.B. generated atomic models; and X.Z., P.Z., T.T.T., S.Y.A., G.D.S., C.R.U., and T.E. generated DNA constructs and performed pharmacological assays. D.W., P.M.S., A.J.K., D.E.G., X.Z., and M.J.B. analyzed the receptor structures; D.W., X.Z., P.Z., C.R.U., and S.R.-R. analyzed pharmacological data. S.G.B.F. generated FLAG resin; P.Ś. synthesized CHU-128. D.W. and P.M.S. supervised the project; A.G. and Y.-L.L. assisted complex biochemistry supervision; and P.Z., S.R.-R., and C.J.L. supervised pharmacological experiments. Project administration was provided by D.W., P.M.S., R.D., and S.R.-R. D.W., X.Z., M.J.B., A.J.K., R.D., and P.M.S. prepared figures and videos. D.W., P.M.S., X.Z., A.J.K., and D.E.G. wrote the manuscript draft. M.J.B., P.Z., T.T.T., S.Y.A., C.R.U., A.C., S.G.B.F., L.J.M., S.R.-R., C.J.L., and R.D. reviewed and edited the manuscript. D.W., P.M.S., R.D., and D.E.G. acquired the financial support for the project.

DECLARATION OF INTERESTS

C.R.U., T.E., and S.R.-R. are employees of Novo Nordisk. P.Ś. is an employee of Apigenex.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
rabbit anti-Gs C-18 antibody	Santa Cruz	Cat# sc-383; RRID: AB_631539
mouse poly-His antibody	QIAGEN	Cat# 34660; RRID: AB_2619735
gp64-PE antibody	Thermo Fisher	Cat# 12-6991-82; RRID: AB_1633369
IRDye 680RD Goat anti-Mouse IgG	LI-COR	Cat# LCR-926-68070: RRID: AB_10956588
IRDye 800CW Goat anti-Rabbit IgG	LI-COR	Cat# LCR-925-32211: RRID: AB_2651127
Chemicals, Peptides, and Recombinant Proteins		
GLP-1(7-36)NH_2	ChinaPeptides	N/A
PF 06882961	Servier, France	N/A
CHU-128	Novo Nordisk, Denmark	N/A
ESF 921 serum-free media	Expression Systems	Cat# 96-001
Opti-MEM I Reduced Serum Medium	Thermo Fisher Scientific	Cat# 31985088
cOmplete Protease Inhibitor Cocktail tablets	Roche	Cat# 5056489001
Apyrase	NEB	N/A
Lauryl maltose neopentyl glycol	Anatrace	Cat# NG310
Cholesteryl hemisuccinate	Anatrace	Cat# CH210
benzonase	Merk Millipore	Cat# 71205-3
Flag peptide	Sigma-Aldrich	Cat# F3290
M1 anti-Flag affinity resin	Sigma-Aldrich	N/A
Superdex 200 Increase 10/300 column	GE Healthcare	Cat# 28990944
Ni-NTA agarose	QIAGEN	Cat# 30230
Anti-Flag M1	ATCC	Hybridoma ATCC HB-259; RRID: CVCL_J730
Amicon Ultra Centrifugal Filter (MWCO, 100 kDa)	Merk Millipore	UFC910024
TGX Precast Gel	BioRad	Cat#4561086
Precision Plus Protein Dual Color Standards	BioRad	Cat# 1610394
Instant Blue	Sigma-Aldrich	Cat# ISB1L
PVDF membrane	BioRad	Cat# 1620264
FuGENE HD Transfection Reagent	Promega	Cat# E2311
Critical Commercial Assays		
Lance cAMP detection kit	PerkinElmer	Cat# AD0264
QuikChange site-directed mutagenesis kit	Agilent	Cat# 200521
Bac to Bac system	Thermo Fisher	Cat# A11098
NanoBRET Nano-Glo substrate	Promega	Cat#N1572
Coelenterazine H	NanoLight	Cat#301
Fluo-8 AM Green Fluorescent Calcium binding dye	Abcam	Cat#142773
Deposited Data		
GLP-1:GLP-1R:Gs coordinates	This manuscript	PDB: 6X18
GLP-1:GLP-1R:Gs EM map	This manuscript	EMD-21992
PF-06882961:GLP-1R:Gs coordinates	This manuscript	PDB: 6X1A
PF-06882961:GLP-1R:Gs EM map	This manuscript	EMD-21994

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
CHU-128:GLP-1R:Gs coordinates	This manuscript	PDB: 6X19
CHU-128:GLP-1R:Gs EM map	This manuscript	EMD-21993
GLP-1:GLP-1R:Gs coordinates (4.1Å)	Zhang et al., 2017	PDB: 5VAI
GLP-1:GLP-1R:Gs EM map (4.1Å)	Zhang et al., 2017	EMD-8653
ExP5:GLP-1R:Gs coordinates	Liang et al., 2018b	PDB: 6B3J
ExP5:GLP-1R:Gs EM map	Liang et al., 2018b	EMD-7039
TT-OAD2:GLP-1R:Gs	Zhao et al., 2020	EMD-20179
β2 adrenergic receptor:Gs coordinates	Rasmussen et al., 2011	PDB: 3SN6
Experimental Models: Cell Lines		
Spodoptera frugiperda	Expression Systems	Cat# 94-001
Trichuplusia ni cells	Expression Systems	Cat# 94-002
CHOFlpIn	Invitrogen	Cat# R75807
Escherichia coli BL21	Thermo Fisher	Cat# C600003
Oligonucleotides		
Primers GLP1R, both insect and mammalian	Liang et al., 2018b	N/A
Recombinant DNA		
pvL1392 dual Gbetagamma	Liang et al., 2018b	N/A
pFastBac-HA-FLAG-GLP1R	Liang et al., 2018b	N/A
pcDNA3.1-HA-FLAG-GLP1R	Liang et al., 2018b	N/A
pFastBac-DNGs	Liang et al., 2020b	N/A
pMESy4-Nb35	Rasmussen et al., 2011	N/A
Software and Algorithms		
MotionCor2	Zheng et al., 2017	https://emcore.ucsf.edu/ucsf-software
GCTF	Zhang, 2016	https://www2.mrc-lmb.cam.ac.uk/research/locally-developed-software/zhang-software/#gctf
crYOLO	Wagner et al., 2019	http://sphire.mpg.de/
RELION (version 3.1)	Zivanov et al., 2018	https://www3.mrc-lmb.cam.ac.uk/relion//index.php?title=Main_Page
CryoSPARC	Punjani et al., 2017	https://cryosparc.com/
UCSF Chimera package	Pettersen et al., 2004	http://www.rbvi.ucsf.edu/chimera/
MDFF	Chan et al., 2012	https://www.ks.uiuc.edu/Research/mdff/
Coot	Emsley et al., 2010	https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/
Phenix software package	Adams et al., 2010	https://www.phenix-online.org/
Dimplot module within the Ligplot+ program	Laskowski and Swindells, 2011	https://www.ebi.ac.uk/thornton-srv/software/LigPlus/
UCSF Chimera package	Pettersen et al., 2004	https://www.cgl.ucsf.edu/chimera/
Prism	GraphPad	https://www.graphpad.com/scientific-software/prism/

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Denise Wootten (denise.wootten@monash.edu).

Materials Availability

All unique/stable reagents generated in this study are available from the Lead Contact with a completed Materials Transfer Agreement.

Data and Code Availability

The atomic coordinates and the cryo-EM density maps generated during this study are available at the protein databank (https://www.rcsb.org) and the electron microscopy databank (https://www.ebi.ac.uk/pdbe/emdb) under PDB accession number 6X18 and EMDB entry ID EMD-21992 for the GLP-1:GLP-1R:G_s structure, PDB accession number 6X1A and EMDB entry ID EMD-21994 for the PF-06882961:GLP-1R:G_s structure, and PDB accession number 6X19 and EMDB entry ID EMD-21993 for the CHU-128:GLP-1R:G_s structure.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Protein expression for biochemical analysis and purification was performed in *Spodoptera frugiperda (Sf9)* and *Trichuplusia ni* (*Tni*) insect cells, maintained in ESF 921 serum free media (Expression systems) at 30°C. CHOFlpIn cells used for mammalian cell studies were maintained in DMEM supplemented with 5% FBS at 37°C in 5% CO_2. *Escherichia coli* (*E. coli*) strain BL21, used to express Nb35, was cultured in Luria-Bertani (LB) liquid medium (10 g tryptone, 10 g NaCL and 5 g yeast extract per liter) with continuous shaking (180 rpm) or on LB agar plate (LB medium with 15 g agar per liter) at 37°C.

METHOD DETAILS

Constructs

The human GLP-1R was modified to replace the native signal peptide with that of haemagglutinin (HA) to improve receptor expression and contain an N-terminal Flag tag and C-terminal 8xHis tag. 3C protease cleavage site (LEVLFQGP) was inserted between both tags and the receptor. The constructs were generated in both mammalian (pcDNA3.1) and insect cell expression (pFastBac) vectors, and the modification did not affect the receptor pharmacological profile (Liang et al., 2018b). Dominant-negative Gαs (DNGαs) that was used has been previously described (Liang et al., 2018c, Liang et al., 2020b). 8xHis tagged Nb35 was provided by B. Kobilka (Rasmussen et al., 2011). Various constructs were used for mammalian cell assays including human GLP-1R in pcDNA3.1 modified with a 2xcMyc-N-terminal epitope tag, and human GLP-1R with a c-terminal Rluc-8 tag. These constructs were validated previously (Fletcher et al., 2018; Koole et al., 2010). For G protein conformational assays, a Nanoluc tag was inserted into Gαs after G(h1ha10) with a SGGGGS linker as described previously (Liang et al., 2018c). This was used in conjunction with Gβ1 and an N-terminally Venus labeled Gγ2.

Insect cell expression

The Bac-to-Bac Baculovirus Expression System (Invitrogen) was used to generate high-titer recombinant baculovirus (> 10^9 viral particles per mL). Recombinant baculovirus was produced by transfecting recombinant bacmids (2-3 μg) into *Sf9* cells (5 mL, density of $4x10^5$ cells per mL) using FuGENE HD Transfection Reagent (Promega) and Opti-MEM Reduced Serum Media (Thermo Fisher Scientific). After 5 d incubation at 27°C, P0 viral stock was harvested as the supernatant of the cell suspension to produce high-titer viral stock (P1 and P2 virus). Viral titers were analyzed by flow cytometry on cells stained with gp64-PE antibody (Expression Systems). Human GLP1R, human DNGαs and human Gβ1-γ2 were co-expressed by infecting *Tni* insect cells at a density of 3.5 million/mL with P2 baculovirus at multiplicity of infection (MOI) ratio of 3:3:1. Culture was harvested by centrifugation 48 h post infection and cell pellets were stored at −80°C.

Purification of Nanobody35

Nanobody-35 (Nb35) bearing a C-terminal His-tag was expressed in the periplasm of *E. coli* strain BL21 following induction with IPTG. Cultures of 1L were grown to OD_{600} to 0.7 at 37°C in TB media containing 0.1% glucose, 2 mM $MgCl_2$ and 100 μg/mL ampicillin. Induced cultures were grown overnight at 25°C. Cells were harvested by centrifugation and lysed in ice-cold buffer (50 mM Tris pH 8.0, 0.125 mM sucrose, 0.125 mM EDTA). Lysate was centrifuged to remove cell debris and Nb35 was purified by nickel affinity chromatography. Elute was concentrated to 5 mg/mL and stored at −80°C.

Complex formation and purification

GLP-1R-DNGs complex formation and purification were performed as described by Liang et al., 2018b. Cell pellets (from 1 L insect cell culture) were thawed and suspended in 20 mM HEPES pH 7.4, 50 mM NaCl, 5 mM $CaCl_2$, 2 mM $MgCl_2$ supplemented with cOmplete Protease Inhibitor Cocktail tablets (Sigma Aldrich) and benzonase (Merk Millipore). Complex was formed by adding GLP-1R agonists (10 μM GLP-1, 50 μM PF 06882961 or 50 μM CHU-128), Nb35–His (10 μg/mL) and apyrase (25 mU/mL, NEB). The suspension was incubated for 1 h at room temperature. The complex was solubilized from membrane using 0.5% (w/v) LMNG and 0.03% (w/v) CHS (Anatrace) for 1 h at 4°C. Insoluble material was removed by centrifugation at 30,000 g for 20 min and the solubilized complex was immobilized by batch binding to M1 anti-Flag affinity resin in the presence of 5 mM $CaCl_2$. The resin was packed into a glass column and washed with 20 column volumes of 20 mM HEPES pH 7.4, 100 mM NaCl, 2 mM $MgCl_2$, 5 mM $CaCl_2$, 1 μM agonist, 0.01% (w/v) LMNG and 0.006% (w/v) CHS followed by elution with buffer containing 5 mM EGTA and 0.1 mg/mL FLAG peptide. The complex was then concentrated using an Amicon Ultra Centrifugal Filter (MWCO, 100 kDa) and subjected to size-exclusion chromatography (SEC) on a Superdex 200 Increase 10/300 column (GE Healthcare) that was pre-equilibrated with 20 mM HEPES pH 7.4,

100 mM NaCl, 2 mM $MgCl_2$, 1 μM agonist, 0.01% (w/v) MNG and 0.006% (w/v) CHS to separate complex from contaminants. Eluted fractions consisting of receptor and G protein complex were pooled and concentrated to 3-5 mg/mL. The complex samples were flash frozen in liquid nitrogen and stored at −80°C.

SDS-PAGE and western blot analysis

Samples from important steps during purification were collected and analyzed by SDS–PAGE and western blot. TGX Precast Gel (BioRad) was used to separate proteins within samples at 200 V for 30 min. Then gels were either stained by Instant Blue (Sigma Aldrich) or immediately transferred to PVDF membrane (BioRad) at 100 V for 1 h. The proteins on the PVDF membrane were probed with two primary antibodies simultaneously, rabbit anti-Gs C-18 antibody (cat. no. sc-383, Santa Cruz) against $G\alpha_s$ subunit and mouse poly-His antibody (cat. no. 34660, QIAGEN) against His tags. The membrane was washed and incubated with secondary anti-mouse and anti-rabbit antibodies (LI-COR). The membranes were imaged on a Typhoon 5 imaging system (Amersham).

Negative staining and data processing

The complex samples were diluted to 0.006 mg/mL in 20 mM HEPES pH 7.4, 100 mM NaCl, 2 mM $MgCl_2$ and 1 μM agonist and applied to the continuous carbon grids (EMS). The grids were stained with 0.8% (w/v) uranyl formate solution and imaged using a cryo-EM Tecnai T12 or L120c TEM at 120 kV. Around 50 images for each complex were collected with a magnified pixel size of 2.06 Å, and ~20,000 particles of each complex were auto picked, extracted and 2D classified by RELION-3.1-beta.

Vitrified sample preparation and data collection

Samples (3 μL) were applied to a glow-discharged UltrAuFoil R1.2/1.3 Au 300 gold foil grids (Quantifoil GmbH, Großlöbichau, Germany) and were flash frozen in liquid ethane using a Vitrobot mark IV (Thermo Fisher Scientific, Waltham, Massachusetts, USA) set at 100% humidity and 4°C for the prep chamber and 10 s blot time. Data were collected on a Titan Krios G3i microscope (Thermo Fisher Scientific) operated at an accelerating voltage of 300 kV with a 50 μm C2 aperture at an indicated magnification of 105K in nanoprobe EFTEM mode. Gatan K3 direct electron detector positioned post a Gatan BioQuantum energy filter (Gatan, Pleasanton, California, USA), operated in a zero-energy-loss mode with a slit width of 25 eV was used to acquire dose fractionated images of the samples with a 100 μm objective aperture. Movies were recorded in hardware-binned mode (previously called counted mode on the K2 camera) with the experimental parameters listed in Table S1 using a 9-position beam-image shift acquisition pattern by custom scripts in SerialEM (Schorb et al., 2019).

Cryo-EM data processing

5739 movies from the GLP-1 dataset were motion corrected using MotionCor2 (Zheng et al., 2017) and subjected to contrast transfer function (CTF) estimation using Kai Zhang's GCTF software (Zhang, 2016). 2.7 million particles were reference-free picked in RELION 3.0 (Zivanov et al., 2018), extracted with a 128 pixels box and subjected to 2D classification. An initial 3D model was generated from selected good 2D classes by Stochastic Gradient Descend (SGD) in RELION and was used for 3D refinement of the particle set. The resulting 3D map, low-pass filtered to 10 Å, was used to reference-based re-pick the micrographs in RELION, producing 3.8 million particles. The re-picked particles were extracted with a 128 pixels box (1.66 Å/pixel) and were subjected to two rounds of 3D classification with 3 classes, each time selecting the best-resolved class. The resulting set of 715K particles was subjected to three rounds of CTF refinement, Bayesian polishing with a larger box/smaller pixel and 3D auto refinement, followed by another round of 3D classification and 3D refinement that produced a 3D map at 2.14 Å in a 320 pixels box (0.83 Å/pixel) from a set of 636K particles. The particle set was then subjected to CTF refinement with higher order aberrations in RELION 3.1 that resulted in the final consensus map at 2.11 Å. The receptor and ECD were further refined with local masks in RELION.

5625 and 5859 movies of PF-06882961- and CHU-128-GLP-1R-Gs complexes, respectively, were motion corrected using MotionCor2 (Zheng et al., 2017) and subjected to CTF estimation with GCTF software (Zhang, 2016). Particles were picked from corrected micrographs using crYOLO software (Wagner et al., 2019). Picked particles (2.36 million and 5.32 million particles of PF 06882961 and CHU-128 complex respectively) were extracted using a box size of 288 pixels and 2D classified using RELION (version 3.1). The selected 2D particles (1.34 million and 2.27 million particles of PF 06882961 and CHU-128 complex respectively) were used to generate the initial 3D model by SGD algorithm for PF 06882961 complex (volume of PF 06882961 complex was used as initial model for CHU-128 complex), and subsequently applied to 3D classification. Particles from the best-looking class (1.21 million and 1.72 million particles of PF 06882961 and CHU-128 complex, respectively) were subjected to Bayesian particle polishing, CTF refinement and cycles of 3D auto-refinement in RELION (Zivanov et al., 2018). The refined particles were subjected to further 3D classification with fine angular sampling. 683K and 934K particles of PF-06882961- and CHU-128-GLP-1R-Gs complexes, respectively, were chosen to generate a final map using 3D auto-refinement and sharpened with a B-factor of $-59Å^2$ and $-25Å^2$, respectively. Local resolution was determined using RELION with half-reconstructions as input maps. The map density of detergent micelle and Gαs helical domain (AHD) was masked out during the consensus map reconstruction for clarity. The agonist-GLP-1R or ECD of each particle was further refined through receptor/ECD-focused 3D auto-refinement with a loose mask in RELION.

To improve the resolution of AHD, the refined particle stack (636K, 732K and 934K particles of GLP-1, PF 06882961 and CHU-128 complexes, respectively) was subjected to 3D classification without alignment into 5 classes using a broad mask of the AHD. Each particle class (52K, 74K, 152K, 255K or 102K of GLP-1 complex; 115K, 116K, 127K, 158K or 216K of PF 06882961 complex; 152K,

181K, 197K, 200K or 203K of CHU-128 complex) was 3D auto-refined, and the best resolved classes (152K, 116K and 181K particles of GLP-1, PF 06882961 and CHU-128 complexes, respectively) were further refined and post processed with a G protein mask to generate a final map of the AHD.

Atomic model refinement

GLP-1-GLP-1R-Gs complex

The sequence corrected model of ExP5-GLP-1R-Gs (PDB: 6B3J) was used as the initial template and fitted in the cryo-EM density maps in UCSF Chimera (v1.14) (Pettersen et al., 2004), followed by molecular dynamics flexible fitting (MDFF) simulation with nanoscale molecular dynamics (NAMD) (Chan et al., 2012). The fitted model was further refined by rounds of manual model building in COOT (Emsley et al., 2010) and real space refinement, as implemented in the PHENIX software package (Adams et al., 2010). The ECD and ECLs were modeled manually without ambiguity based on the ECD-focused map. Density of ECD linker ($K130^{ECD}$-$S135^{ECD}$) and ICL3 ($L339^{ICL3}$-$T343^{ICL3}$) was not observed and these residues were omitted from the final model.

PF 06882961 or CHU-128-GLP-1R-Gs complex

Geometry restraint information of PF 06882961 and CHU-128 was generated using the module 'electronic Ligand Builder and Optimization Workbench (eLBOW)' in PHENIX (Moriarty et al., 2009). The model of GLP-1- GLP-1R-Gs or PF-06882961-GLP-1R-Gs was used as initial template for PF 06882961 or CHU-128-GLP-1R-Gs complex modeling, respectively. The template was fit into the cryo-EM density map with the MDFF routine in NAMD (Chan et al., 2012). The fitted models were further refined by manual model building in COOT and real space refinement in PHENIX based on the consensus maps and receptor-focused maps (Adams et al., 2010; Emsley et al., 2010). The model of ECD in the CHU-128-complex was further optimized according to the ECD-focused map. Cryo-EM density for residues between $S129^{ECD}$ and $R134^{ECD}$ in the ECD linker region of both small molecule complexes were discontinuous and these sequences were omitted from the final model. The ECL3 was less resolved in CHU-128-complex and residues from $A375^{ECL3}$-$G377^{ECL3}$ were omitted.

The AHD of the three complexes was modeled based on their AHD refined maps using the AHD of β2 adrenergic receptor-Gs complex (PDB: 3SN6) as the initial template. Water molecules were modeled manually into clear point density for all the three structures and further refined in Coot and PHENIX (Adams et al., 2010; Emsley et al., 2010). The final models were subjected to global refinement and comprehensive validation implemented in PHENIX. The cryo-EM data collection, refinement and validation statistics for all complexes are reported in Table S1.

Model residue interaction analysis

Interactions in the PDB files between the bound agonist and the receptor were analyzed using the "Dimplot" module within the Ligplot+ program (v2.2) (Laskowski and Swindells, 2011). Hydrogen bonds were additionally analyzed using the UCSF Chimera package (Pettersen et al., 2004), with relaxed distance and angle criteria (0.4 A and 20 degree tolerance, respectively).

Cryo EM dynamics analysis

3D variability analysis implemented in cryoSPARC (v2.9) (Punjani et al., 2017) was performed to understand and visualize the dynamics in GLP-1R complexes, as previously described for analysis of the dynamics of adrenomedullin receptors (Liang et al., 2020a). The final polished particle stack including GLP-1, PF 06882961 and CHU-128-GLP-1R-Gs complexes from RELION consensus refinement were imported into the cryoSPARC environment. 2D classification and selection were initially performed to ensure only highly resolved particles were further analyzed. These particles were then subjected to 3D refinement, using a low pass filtered RELION consensus map as an initial model and a generous mask created in cryoSPARC as default. 3D variability of these GLP-1R complexes was analyzed in 3 motions and the 20 volume frame data in each motion was generated in cryoSPARC (Punjani et al., 2017). Output files were visualized in UCSF ChimeraX volume series and captured as movies (Goddard et al., 2018).

Stable cell lines generation

The wild-type (WT) and mutant GLP-1R constructs containing designed signal alanine mutation were integrated into CHOFlpIn cells using FlpIn Gateway technology system (Invitrogen). Stable CHOFlpIn expression cell lines were selected using 600 μg/mL hygromycin B, and maintained in DMEM supplemented with 5% (V/V) FBS (Invitrogen) at 37°C in 5% CO_2.

cAMP accumulation assay – stable cell lines

HEK293A WT GLP-1R, CHOFlpIn WT GLP-1R or CHOFlpIn mutant GLP-1R cells were seeded at a density of 30,000 cells per well into a 96-well plate and incubated overnight at 37°C in 5% CO_2. cAMP detection was using a Lance cAMP kit (PerkinElmer Life and Analytical Sciences), performed as previously described (Hager et al., 2017). Growth media was replaced with stimulation buffer [phenol-free DMEM containing 0.1% (w/v) ovalbumin and 1 mM 3-isobutyl-1-methylxanthine] and incubated for 1 h at 37°C in 5% CO_2. Cells were stimulated with increasing concentrations of ligand, 100 μM forskolin or vehicle, and incubated for 30 min at 37°C in 5% CO_2. The reaction was terminated by rapid removal of the ligand-containing buffer and addition of 50 μL of ice-cold 100% ethanol. After ethanol evaporation, 75 μL of lysis buffer [0.1% (w/v) BSA, 0.3% (v/v) Tween 20, and 5 mM HEPES, pH 7.4] was added, and 10 μL of lysate was transferred to a 384-well OptiPlate (PerkinElmer Life and Analytical Sciences). 5 μL of 1/100 dilution of the Alexa Fluor® 647-anti cAMP antibody solution in the Detection Buffer (PerkinElmer Life and Analytical Sciences) and 10 μL

of 1/5000 dilution of Eu-W8044 labeled streptavidin and Biotin-cAMP in the Detection Buffer (PerkinElmer Life and Analytical Sciences) were added in reduced lighting conditions. Plates were incubated at room temperature for 2h before measurement of the fluorescence using an EnVision Multimode Plate (Reader PerkinElmer Life and Analytical Sciences). All values were converted to cAMP concentration using cAMP standard curve performed parallel and data were subsequently normalized to the response of 100 μM forskolin in each cell line, and then normalized to the WT for each agonist.

Calcium mobilization

HEK293A cells stably expressing the hGLP-1R were seeded at a density of 30,000 cells/well into poly-D-Lysine coated 96-well culture plates and incubated overnight at 37°C in 5% CO_2. Ca^{2+} mobilization was assessed as previously described (Koole et al., 2010). Briefly, media was changed and cells were incubated with fluo8 for 1 h. Ligand-mediated intracellular calcium mobilization was determined using a FlexStation (Molecular Devices, USA), with excitation wavelength at 485 nm and emission wavelength at 525 nm every 1.5 s for 2 minutes, including a 15 s baseline with addition of agonists at increasing concentrations.

pERK1/2 assays

HEK293 cells stably expressing hGLP-1R were seeded at a density of 30,000 cells/well into Poly-D-Lysine coated 96-well culture plates and incubated overnight at 37°C in 5% CO_2. Receptor mediated pERK1/2 was determined using the AlphaScreen ERK1/2 SureFire protocol as previously described previously (Wootten et al., 2013a), with increasing concentrations of the compound. Time course experiments were performed using a single ligand concentration to determine the peak response time point (3min) for the subsequent concentration response curves as previously described.

beta arrestin1/2 assays

HEK293A Cells were transiently transfected with GLP-1R-Rluc8 and rat β-arrestin1-Venus at a ratio of 1:4 using a standard PEI protocol (6:1 ratio PEI:DNA) and seeded in to poly-D-Lysine coated white 96-well culture plates at a density of 30,000 cells/well. Plates were incubated at 37°C in 5% CO_2. 48 h post transfection, β-arrestin recruitment was measured in assay buffer (Phenol red-free DMEM, 0.1% (w/v) ovalbumin) as previously described (Savage et al., 2013). The concentration response curves were plotted using the total area under the curve during the time of measurement post ligand addition.

FYVE trafficking

HEK293A cells were transfected with GLP-1R-Rluc8 and early endosomal marker FYVE tagged with rGFP at 1:4 ratio using a standard PEI protocol (6:1 ratio PEI:DNA) and seeded at a density of 30,000 cells/well in poly-D-Lysine coated white 96-well plate. Following 48 h incubation at 37°C in 5% CO_2, ligand-mediated receptor trafficking was measured in the presence of 2 μM prolume purple substrate in assay buffer (Phenol red-free DMEM, 0.1% (w/v) ovalbumin) as described previously. Plates were read using the PHERAstar FS plate reader (BMG Labtech, Ortenberg, Germany) two wavelengths (emission 1: 410-80nm and emission 2: 515-30nm) at 2 min intervals for 17 min, including a 2-cycle baseline reading preceding compound addition. The concentration response curves were then plotted using the total area under the curve during the time of measurement post ligand addition.

G_s conformational change

HEK293AΔS/Q/12/13 cells stably expressing the GLP-1R (tested and confirmed to be free from mycoplasma) were transfected with a 1:1:1 ratio of $G\gamma_2$:venus–$G\alpha_s$:nanoluc–$G\beta_1$ using a standard PEI protocol (6:1 ratio PEI:DNA). Cells were incubated overnight at 37°C in 5% CO_2. Collection and preparation of cell plasma membranes and assessment of G protein conformational change was performed using a previously established method (Furness et al., 2016). Briefly, 5 μg per well of cell membrane was incubated with furimazine (1:1,000 dilution from stock) in assay buffer (1 × HBSS, 10 mM HEPES, 0.1% (w/v) ovalbumin, 1 × P8340 protease inhibitor cocktail, 1 mM DTT and 0.1 mM PMSF, pH 7.4). The GLP-1R-induced BRET signal between $G\alpha_s$ and Gγ was measured at 30°C using a PHERAstar (BMG LabTech). Baseline BRET measurements were taken for 2 min before addition of vehicle or ligand. BRET was measured at 15 s intervals for a further 10 min. All assays were performed in a final volume of 100 μL. The concentration response curves were then plotted using the total area under the curve during the time of measurement post ligand addition.

NanoBRET kinetic binding assays

HEK293A cells were transiently transfected with Nluc-hGLP-1R. 48 h post transfection, cells were harvested and plasma membrane was extracted as described previously (Furness et al., 2016). 1 μg per well of cell membrane was incubated with furimazine (1:1,000 dilution from stock) in assay buffer (1 × HBSS, 10 mM HEPES, 0.1% (w/v) OVA, 1 × P8340 protease inhibitor cocktail, 1 mM DTT and 0.1 mM PMSF, pH 7.4). AF568-GLP-1 was used as the fluorescent ligand in the NanoBRET binding assay. Membranes were pre-incubated with PF 06882961 or CHU-128 for 10 min prior to assessment of BRET signal between Nluc-hGLP-1R and AF568-GLP-1 This was assessed using a PHERAstar (BMG LabTech) at 10 s intervals (25°C). A 2 min baseline reading was taken before addition of test compound, AF568-GLP-1 (Kd concentration, 3.16 nM, determined previously) was added 10 min later, and the measurement continued for 20 min. Data were corrected for baseline and vehicle responses.

Species selectivity data

Transient transfections

Mammalian expression plasmids encoding either human GLP-1R, mouse GLP-1R, human GLP-1R-W33S or mouse GLP-1R-S33W were purchased from ThermoFisher Scientific. Suspension CHO cells were seeded at 0.2-0.3 mill/mL in shaker flasks 2-3 days prior to transfection and cells were propagated to 1-2 mill/mL at 37°C until the day of transfection. Cells were electroporated separately with the 4 different human and mouse GLP-1R expression plasmids and the resulting transfected cells were further cultivated at 37°C for 24 h. One day post-electroporation, cells were subjected to a temperature drop to 32°C, and cells were further cultivated at 32°C for another 24 h until harvest.

cAMP assays

Ligand-induced accumulation of cAMP was quantified using the cAMP dynamic Gs kit (Cisbio). Cells were pelleted and washed once in Dulbecco's Phosphate Buffered Saline (DPBS, ThermoFisher Scientific). Cells were counted, pelleted and resuspended in assay buffer (Hanks' balanced salt solution buffer (ThermoFisher Scientific), 20 mM HEPES pH 7.4, 1 mM $CaCl_2$, 1 mM $MgCl_2$, 0.1% Pluronic F-68) to a cell density of 0.5 million cells/mL. Cells (10 μl) were stimulated with increasing concentrations of GLP-1, CHU-128 or PF 06882961 (10 μl) for 30 min at 37°C. The assay was stopped by cell lysis and the response was measured by adding 20 μl/well detection solution (cAMP conjugate and lysis buffer (Cisbio) + 2.5% anti-cAMP cryptate conjugate (Cisbio) + 2.5% cAMP-d2 conjugate (Cisbio). The plate was incubated in the dark for 60 min at room temperature (~22°C) and subsequently read on an EnVision Multilabel Reader (PerkinElmer Life and Analytical Sciences), with excitation at 340 nm and measurements of emission at 615 and 665 nm. The fluorescence resonance energy transfer ratios (665/615 nm) were converted to cAMP concentrations by interpolating values from a cAMP standard curve.

Data analysis

All pharmacological data were analyzed using Prism 8.0. Concentration response data were generated by applying a three-parameter logistic equation to either fixed time measurements or data extracted from temporal traces. To quantify kinetic parameters, a one phase association equation was used. Biased agonism was quantified by applying the operational model of agonism to concentration response data where the equation was revised such that the transduction ratio τ/K_A was directly estimated (Kenakin et al., 2012) . Changes to τ/K_A were established by comparing compounds to the reference ligand, GLP-1, and reference pathway, cAMP production.

Graphics

Molecular graphics images were produced using the UCSF Chimera (v1.14) and ChimeraX packages from the Computer Graphics Laboratory, University of California, San Francisco (supported by NIH P41 RR-01081 and R01-GM129325) (Goddard et al., 2018; Pettersen et al., 2004).

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Supplemental Information

Differential GLP-1R Binding and Activation by Peptide and Non-peptide Agonists

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Supplemental Information

Table S1. CryoEM and model parameters. *Related to Figures 1 and S2*

Data Collection	*GLP-1-GLP-1R-Gs*	*PF-06882961-GLP-1R-Gs*	*CHU-128-GLP-1R-Gs*
Magnification	105,000	105,000	105,000
Voltage (kV)	300	300	300
Spot size	4	4	5
Electron exposure (e-/Å^2)	65.4	60.1	54.6
Exposure time (s)	3	3	5
Movie frames	75	75	71
K3 CDS mode	No	No	Yes
Defocus range (μm)	0.6-1.4	0.5-1.3	0.7-1.5
Pixel size (Å)	0.826	0.826	0.826
Symmetry imposed	C1	C1	C1
Final particle imaged (no.)	635,785	683,444	934,493
Resolution (Å)	2.1 (2.2[a], 2.4[b], 2.4[c])	2.5 (2.5[a], 2.8[b], 2.2[c])	2.1 (2.2[a], 2.3[b], 2.8[c])
FSC threshold	0.143	0.143	0.143
Refinement			
Initial model used (PDB code)	6B3J	6X18	6X1A
Map sharpening B factor (Å^2)	-44	-59	-25
Model composition			
Non-hydrogen atoms	10371	10140	10277
Protein residues	1289	1262	1258
ligand	0	1	1
RMSDs			

Bond length (Å)	0.007	0.006	0.007
Bond angles (°)	1.106	0.888	1.032
Validation			
MolProbity score	1.74	2.35	1.75
Clashscore	10.83	13.98	12.45
Rotamer outliers (%)	1.45	3.79	1.01
Ramachandran plot			
Favoured (%)	97.71	96.15	97.26
Allowed (%)	2.21	3.85	2.66
Disallowed (%)	0.08	0	0.08

[a]Resolution (Å) of receptor-focused cryo-EM map

[b]Resolution (Å) of ECD-focused cryo-EM map

[c]Resolution (Å) of complex including the "down" conformation of the $G\alpha_s$ AHD

Table S2: Interactions between the GLP-1R and GLP-1, PF-06882961 or CHU-128.
Related to Figures 2 and S4

GLP-1R Receptor residues		GLP-1	PF-06882961	CHU-128
ECD	V30^{ECD}	E21		
	S31^{ECD}	E21	Pyridine	Methyl pyrazolopiperidine
	L32^{ECD}	E21 (H bond) A24 F28	Fluorobenzyl group Ethoxy linker	
	W33^{ECD}		4-cyano-2-fluorobenzyl group (π-π interaction) Pyridine (π-π interaction)	Indole (π-π interaction) Dimethylmorpholine
	E34^{ECD}			Methyl pyrazolopiperidine
	T35^{ECD}	F28 I29		
	V36^{ECD}		4-cyano-2-fluorobenzyl group	
	Q37^{ECD}		4-cyano group	
	W39^{ECD}	F28 (π-π interaction) L32 R36 (π-π interaction)		
	E68^{ECD}	L32 R36		
	Y69^{ECD}	I29 L32 V33		
	P90^{ECD}	I29		
	W91^{ECD}	I29		
	R121^{ECD}	V33 (H bond)		
	E128^{ECD}	K26		
TM1	P137$^{1.32}$	Y19		Methyl pyrazolopiperidine Dimethyl fluorophenyl group Imidazolone
	E138$^{1.33}$	Y19		Dimethyl fluorophenyl group Methyl fluoroindazole
	L141$^{1.36}$	F12	Pyridine	Dimethyl fluorophenyl group
	L144$^{1.39}$	F12		Dimethyl fluorophenyl group
	Y145$^{1.40}$			Dimethyl fluorophenyl group Methyl fluoroindazole (π-π interaction)
	Y148$^{1.43}$	F12 (π-π interaction)	*GLP-1R F385$^{7.40}$ (π-π interaction)*	Dimethyl fluorophenyl group

				(π-π interaction)
	Y152$^{1.47}$	E9 (H bond)	Carboxylic acid (water mediated)	
TM2	R190$^{2.60}$	E9 (H bond)	Carboxylic acid (water mediated)	
	K197$^{2.67}$	T13 (H bond)	Benzimidazole (H bond)	Imidazolone Indole
	D198$^{2.68}$			Methyl fluoroindazole Imidazolone
	L201$^{2.71}$	V16	Piperidine	Dimethylmorpholine
	K202 $^{2.72}$			Methyl fluoroindazole
	W203$^{2.73}$		Piperidine	
	Y205$^{2.75}$	S17 (H bond) L20 E21 (H bond)		Methyl fluoroindazole (π-π interaction)
ECL1	S206^{ECL1}		Fluorobenzyl-oxy group	
	T207^{ECL1}		Fluorobenzyl-oxy group	
	Q210^{ECL1}	A24 E27		
	W214^{ECL1}	F28 (π-π interaction) W31 (π-π interaction)		
	L217^{ECL1}		4-cyano-2-fluorobenzyl group	
	Y220^{ECL1}			Dimethylmorpholine (H bond)
	Q221^{ECL1}	E21	4-cyano group Oxetane group	
TM3	C226$^{3.29}$			Dimethylmorpholine
	V229$^{3.32}$			Dimethylmorpholine
	F230$^{3.33}$	T13	Oxetane group Benzimidazole-6-carboxylic acid (π-π interaction)	Dimethylmorpholine Indole (π-π interaction)
	M233$^{3.36}$	T13	Benzimidazole	Dimethylmorpholine
	Q234$^{3.37}$	H7 (H bond)	Carboxylic acid (water mediated)	
	V237$^{3.40}$	H7		
	Y241$^{3.44}$	A8 (water mediated) E9 (water mediated)		
ECL2	C296^{ECL2}		Oxetane group	
	T298^{ECL2}	G10 (water mediated) T13 S14 S17 (H bond)	4-cyano group	Indole Methylcyclopropyl group Carbonyl group (water mediated)
	R299^{ECL2}	S14 S17 (H bond) S18	Carboxylic acid (water mediated)	

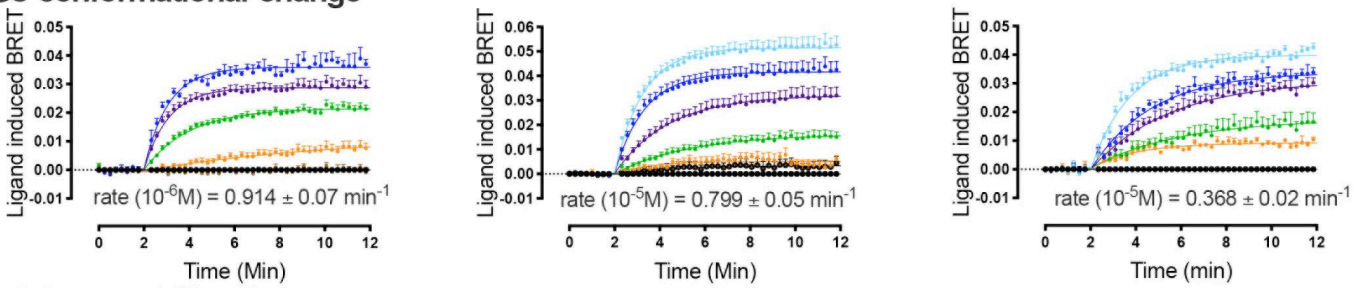
		E21 (H bond)		
	$N300^{ECL2}$	G10 S14 (H bond)		Methylcyclopropyl group
TM5	$W306^{5.36}$	H7 (π-π interaction), G10, T11		
	$I309^{5.39}$	H7		
	$R310^{5.40}$	H7 (water mediated)		
	$I313^{5.43}$	H7		
ECL3	$D372^{ECL3}$	T11		
TM7	$R380^{7.35}$	T11 D15 (ionic interaction)	Carboxylic acid (ionic interaction)	
	$F381^{7.36}$		Piperidine Benzimidazole (π-π interaction)	
	$L384^{7.39}$	A8 T11 D15	Benzimidazole	
	$F385^{7.40}$		Piperidine	
	$E387^{7.42}$	A8		
	$L388^{7.43}$	A8 E9 (water mediated), F12		Dimethyl fluorophenyl group

Receptor residues within 4 Å of the bound ligand (GLP-1, PF-06882961 or CHU-128) are listed. Hydrogen (H) bonds were determined using Ligplot+ and UCSF Chimera. Residues in blue are interactions not shown in Ligplot+.

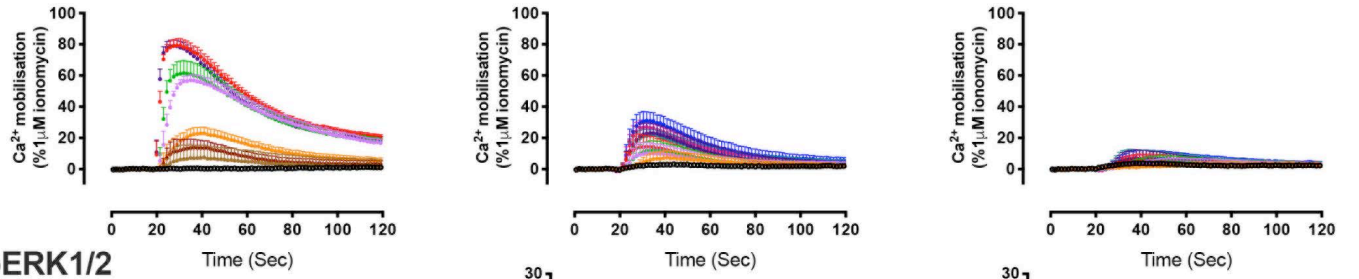
Figure S1. Kinetic profiles of binding, transducer coupling and signalling mediated by GLP-1 and non-peptidic agonists. *Related to Figure 1.* **A,** Agonist-induced changes in trimeric G_s protein conformation measured as a ligand-induced change in BRET between Nluc-$G\alpha_s$ and Venus-$G\gamma$ in membranes prepared from HEK293 cells expressing the GLP-1R. Rates were calculated using a one phase association model; the rate reported was from an approx. EC_{90} concentration of each agonist. **B,** Kinetics of calcium signalling in live HEK293 cells expressing the GLP-1R. **C,** A time-course for pERK1/2 was performed at a single ligand concentration to determine the peak response. This revealed similar profiles of pERK1/2 kinetics, albeit with different maximal signals. **D-E**, Kinetics of β-arrestin-1 (D) and β-arrestin-2 recruitment (E). **F,** Kinetics for ligand-induced BRET between GLP-1R-Rluc8 and the early endosomal marker FYVE-Venus**.** Rates were determined using a one phase association model; the rate reported was from an approx. EC_{90} concentration of each agonist. **G,** Kinetic ligand binding assay in membranes measuring nBRET between Nluc-GLP-1R and the fluorescent probe AF568-GLP-1 in the presence and absence of increasing concentrations of unlabelled agonists. The colour key at the bottom provides information regarding the ligand concentrations used for each ligand in each assay. All experiments were performed in GLP-1R HEK293 cells and with the exception of panel G (n=3), all panels show the mean + S.E.M of 5 independent experiments performed in duplicate or triplicate.

Figure S1. GLP-1 PF 06882961 CHU-128

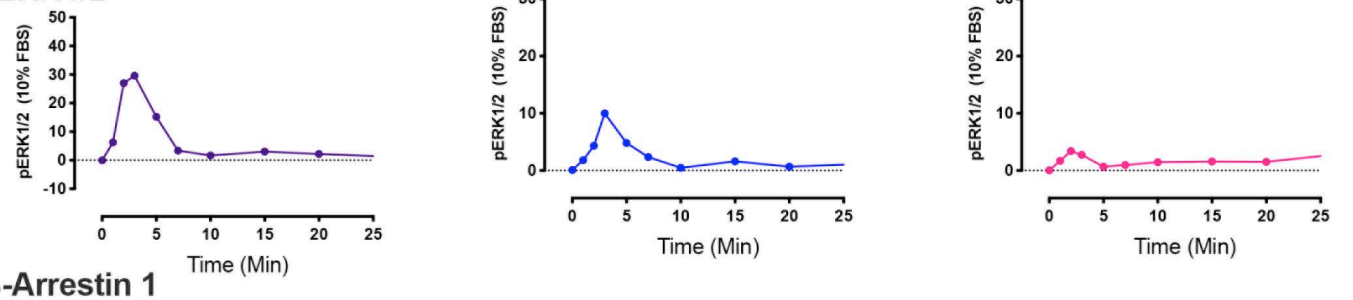
A . Gs conformational change



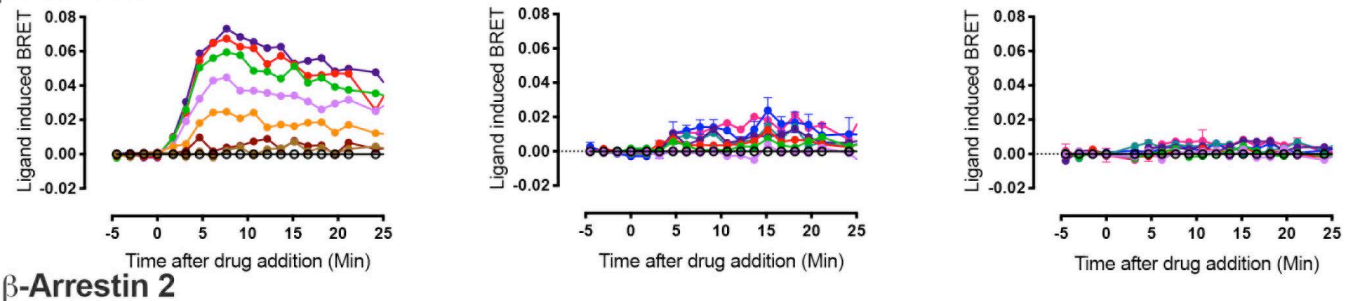
B. Calcium mobilisation



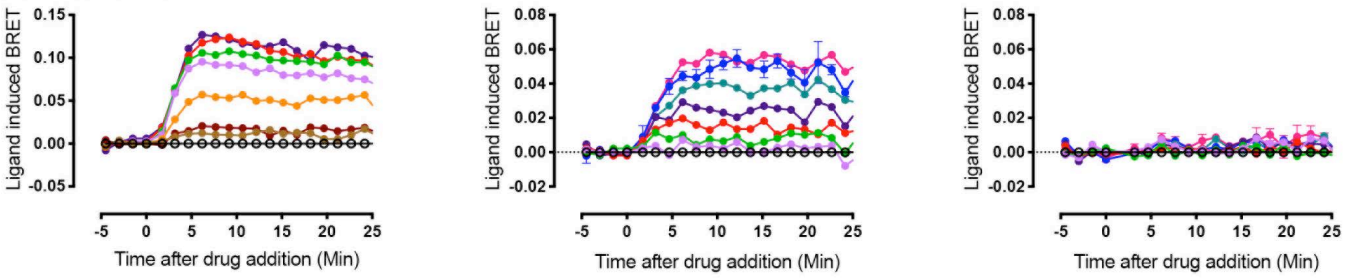
C. pERK1/2



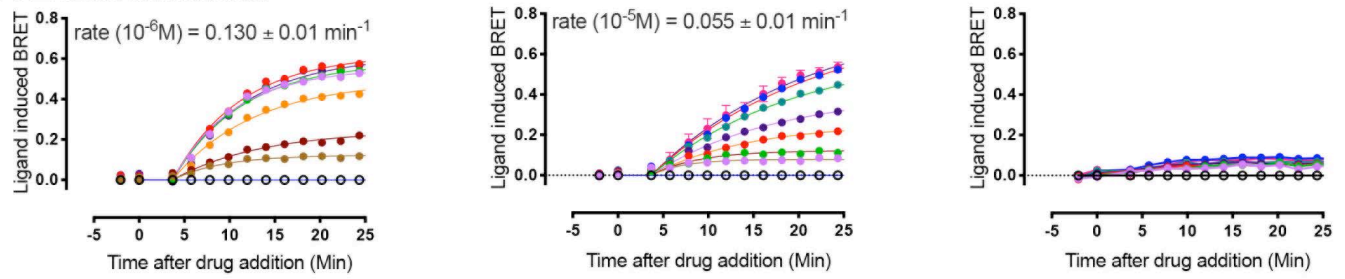
D. β-Arrestin 1



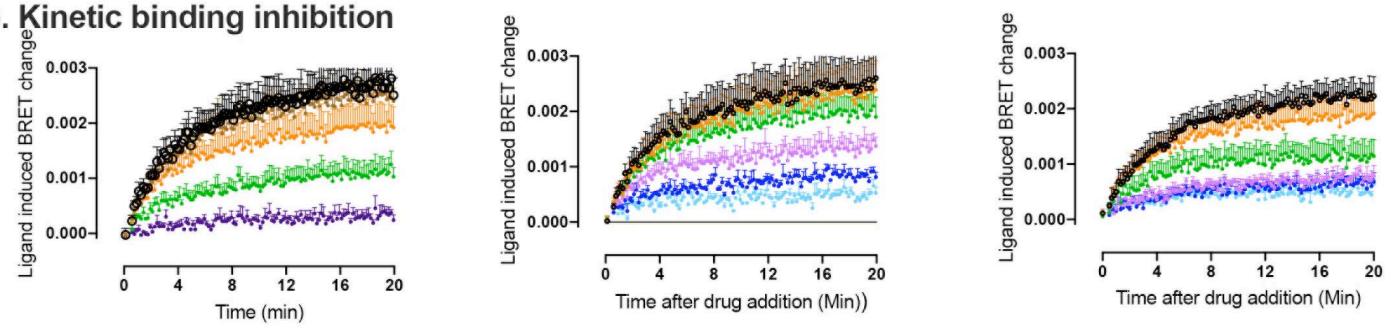
E. β-Arrestin 2



F. FYVE colocalisation



G. Kinetic binding inhibition



Ligand Concentrations: 1nM 3nM. 10nM. 30nM. 100nM. 300nM. 1μM. 3μM. 10μM. 30μM. 100μM

Figure S2. Purification, cryo-EM data imaging and processing of GLP-1R:G_s complexes. *Related to Figures 2 and 3*. **Top,** GLP-1 bound complex; **Middle**, PF 06882961 bound complex; **Bottom,** CHU-128 bound complex. Within each panel: Top left, size exclusion chromatography profile and Coomassie gel of the purified complex. Top middle, 3-D histogram representation of the Euler angle distribution of all the particles used in the reconstruction overlaid on the density map drawn on the same coordinate axis. Top right; exemplar micrograph and 2D class averages of cryo-EM projections of the complex. Bottom left, Gold standard Fourier shell correlation (FSC) curves for the final consensus maps and map validation from half maps, showing the overall nominal resolution. Bottom middle-right, Local resolution-filtered EM maps (consensus and receptor/ECD focused refinements) displaying local resolution (Å) coloured from highest resolution (dark blue) to lowest resolution (red).

Figure S2

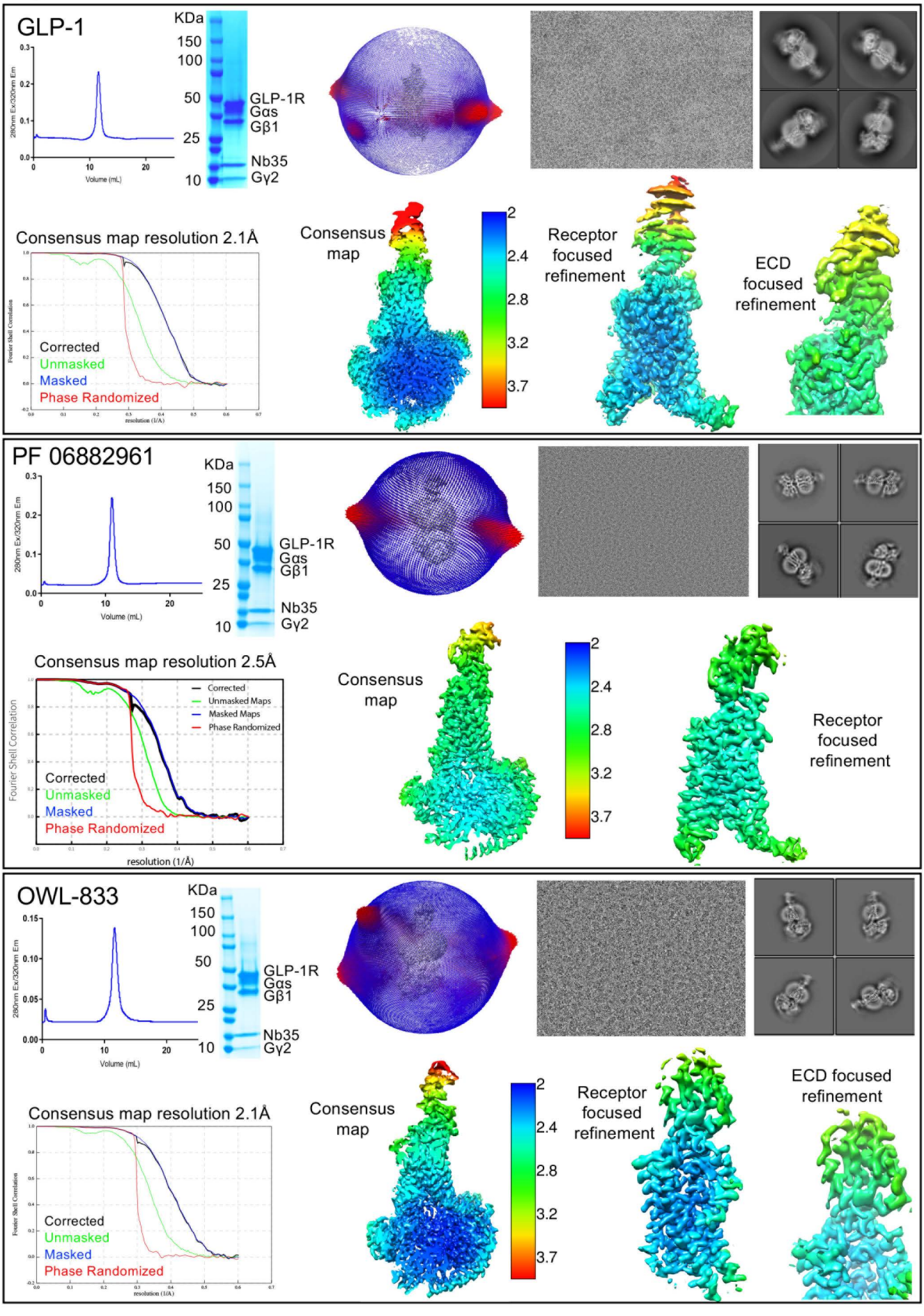

GLP-1
KDa
150
100
50
25
10
GLP-1R
Gαs
Gβ1
Nb35
Gγ2
Volume (mL)
Consensus map resolution 2.1Å
Fourier Shell Correlation
Corrected
Unmasked
Masked
Phase Randomized
resolution (1/Å)
Consensus map
2
2.4
2.8
3.2
3.7
Receptor focused refinement
ECD focused refinement
PF 06882961
KDa
150
100
50
25
10
GLP-1R
Gαs
Gβ1
Nb35
Gγ2
Volume (mL)
Consensus map resolution 2.5Å
Fourier Shell Correlation
Corrected
Unmasked Maps
Masked Maps
Phase Randomized
Corrected
Unmasked
Masked
Phase Randomized
resolution (1/Å)
Consensus map
2
2.4
2.8
3.2
3.7
Receptor focused refinement
OWL-833
KDa
150
100
50
25
10
GLP-1R
Gαs
Gβ1
Nb35
Gγ2
Volume (mL)
Consensus map resolution 2.1Å
Fourier Shell Correlation
Corrected
Unmasked
Masked
Phase Randomized
resolution (1/Å)
Consensus map
2
2.4
2.8
3.2
3.7
Receptor focused refinement
ECD focused refinement

Figure S3. Atomic models of the ligands, receptors and Gα$_s$ α5 helix in the cryo-EM map. *Related to Figures 2, 3, 6 & S7*. EM density map and the model are shown for GLP-1, PF 06882961, CHU-128 and all seven TM helices, and ECLs of the GLP-1R when bound to each agonist. The α5 helix of the Gα$_S$ Ras-like domain is also shown for each complex. For GLP-1R, the consensus map was used, with the exception of those labelled * where the ECD refined map is shown. For PF 06882961- and CHU-128 receptor and ligand density was from the receptor-focused refined maps. For the α5 helix, all density shown is from the consensus cryo-EM maps for the respective complex. ECL3 for CHU-128 was poorly resolved and was not modelled.

Figure S3

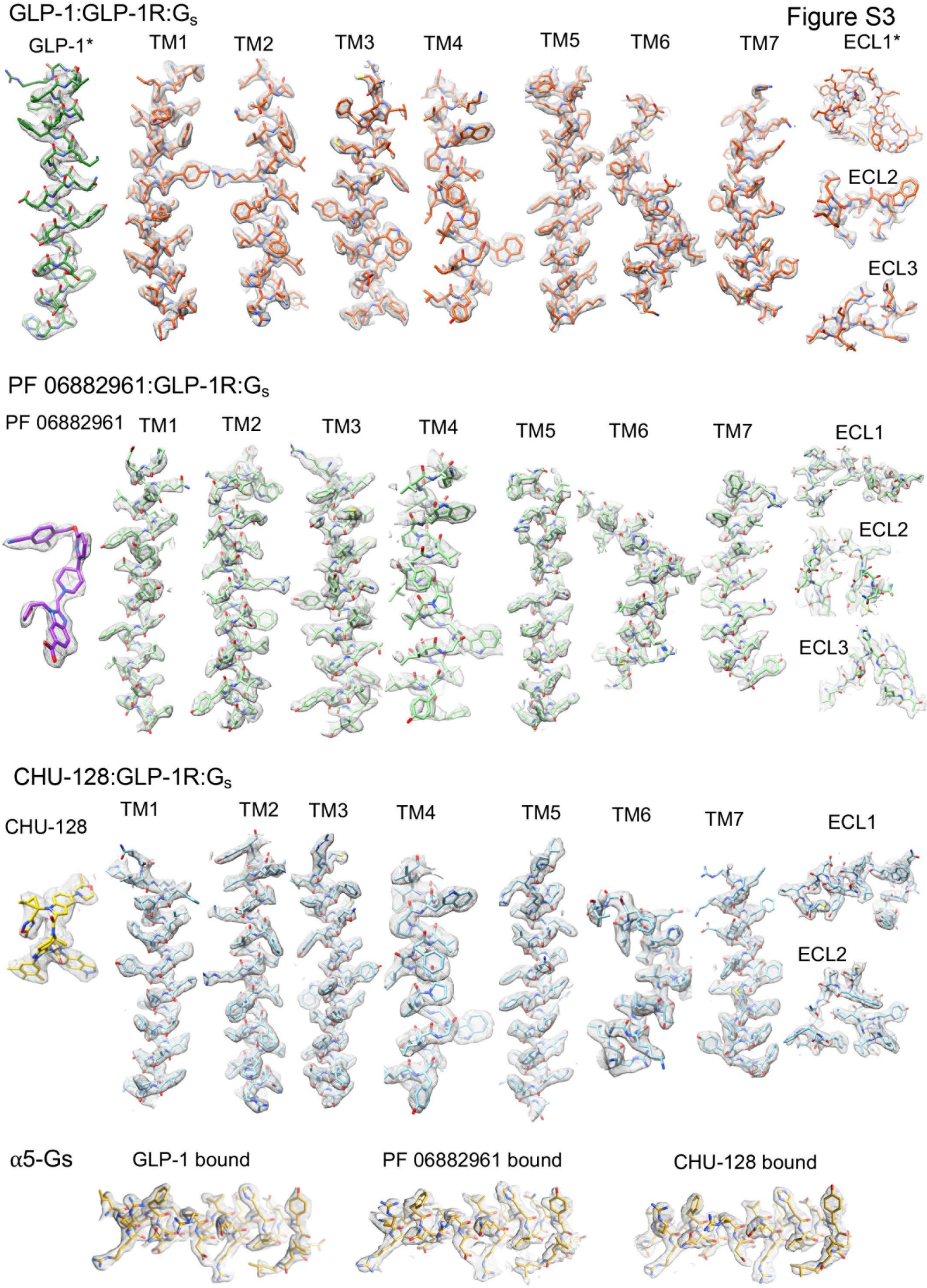
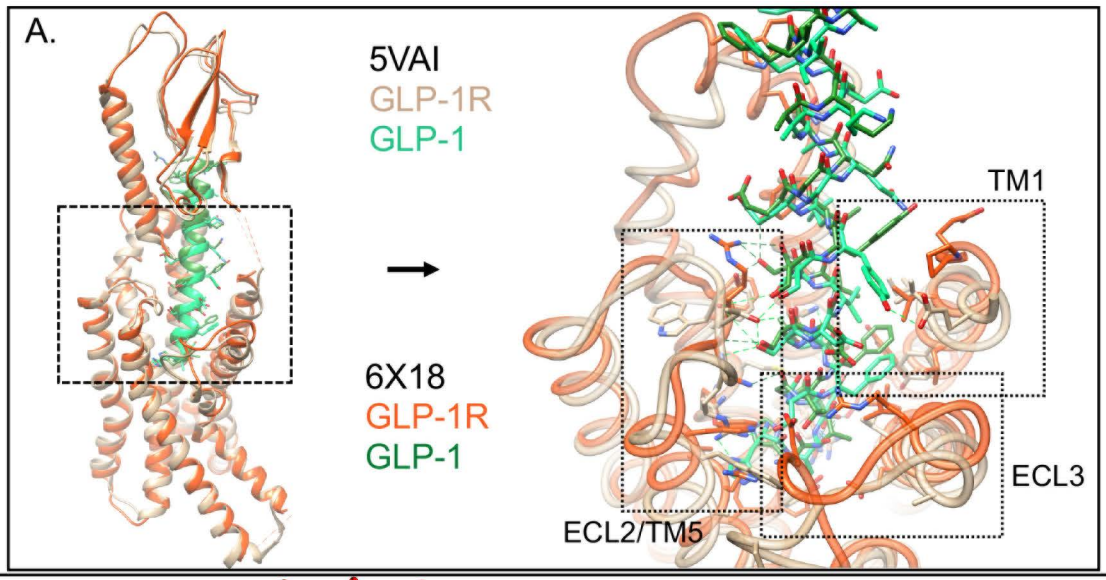

GLP-1:GLP-1R:Gs
GLP-1*
TM1
TM2
TM3
TM4
TM5
TM6
TM7
ECL1*
ECL2
ECL3
PF 06882961:GLP-1R:Gs
PF 06882961
TM1
TM2
TM3
TM4
TM5
TM6
TM7
ECL1
ECL2
ECL3
CHU-128:GLP-1R:Gs
CHU-128
TM1
TM2
TM3
TM4
TM5
TM6
TM7
ECL1
ECL2
α5-Gs
GLP-1 bound
PF 06882961 bound
CHU-128 bound

Figure S4. Ligand interactions with the GLP-1R. *Related to Figure 3.* **A,** Overlay of 5VAI with the 2.1 Å GLP-1 bound GLP-1R structure reveals differences within the extracellular region between the two structures. These reveal differences in modelled interactions between GLP-1 and residues within TM1, ECL2/TM5 and TM6/ECL3/TM7. **B,** Top, interactions between the N-terminal portion of GLP-1 (residues 7-17) and the GLP-1R TM bundle as determined by Ligplot+. Middle; Atomic models of the binding sites of the PF-06882961-, GLP-1- and CHU-128- bound GLP-1Rs and their density within the cryo-EM maps. Bottom, interactions between PF 06882961 (left) and CHU-128 (right) with the GLP-1R as determined by Ligplot+. In the Ligplot+ figures, hydrogen bonds are shown as green dotted lines, whereas red dotted lines are hydrophobic contacts. Blue spheres indicate waters that directly interact with ligands.

Figure S4


A.
5VAI
GLP-1R
GLP-1
6X18
GLP-1R
GLP-1
TM1
ECL3
ECL2/TM5

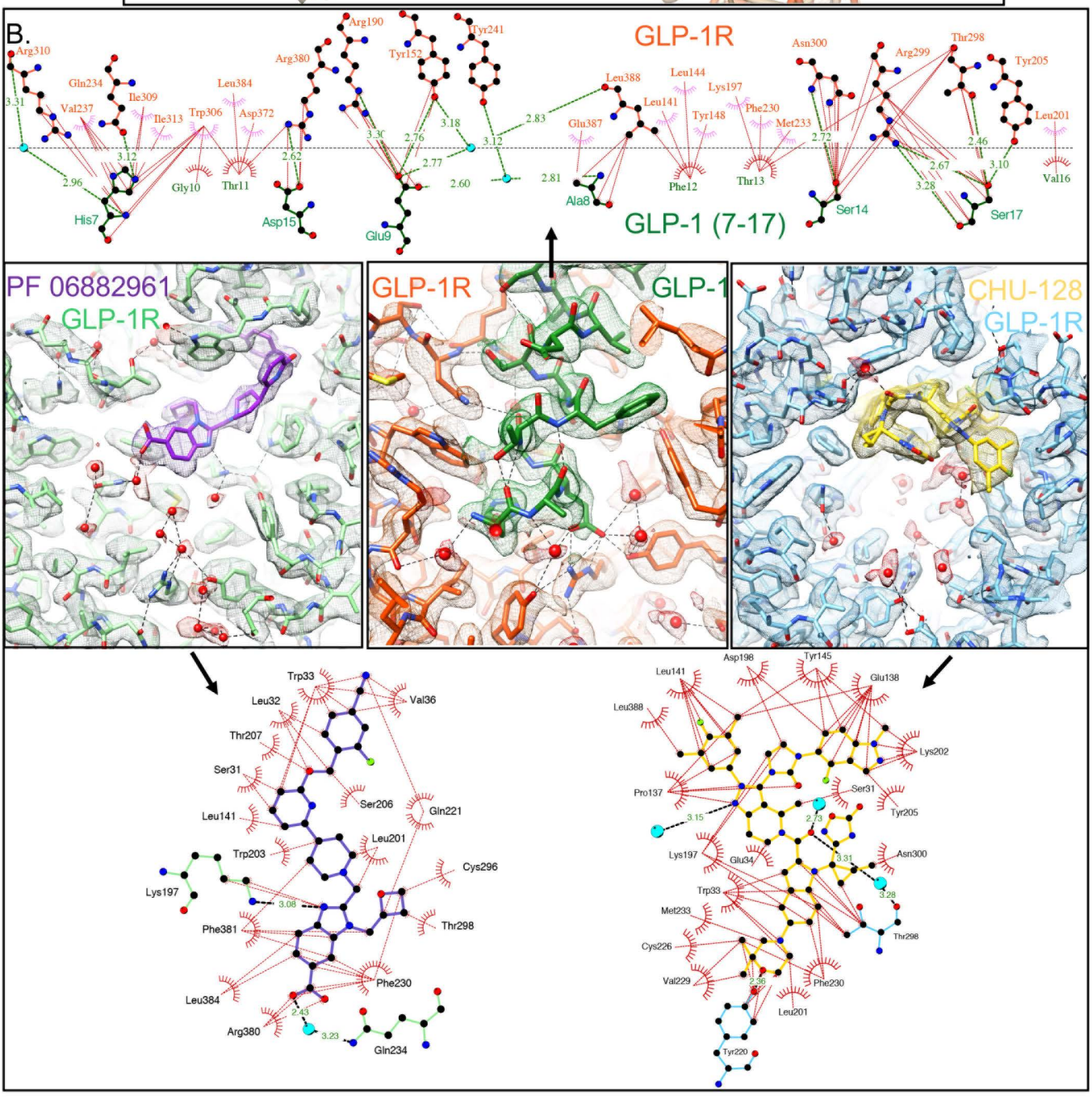

B.
GLP-1R
GLP-1 (7-17)
PF 06882961
GLP-1R
GLP-1R
GLP-1
CHU-128
GLP-1R

Figure S5. Ligand mediated cAMP accumulation in ChoFlpIn cells stably expressing WT or mutant hGLP-1Rs. *Related to Figures 3, 4 and 6.* Concentration response curves for alanine mutants of the GLP-1R were assessed for GLP-1, PF 06882961 and CHU-128. Residues that either interacted with PF 06882961, CHU-128 or both agonists were selected to assess their importance in ligand-mediated receptor activity. A select set of residues that interact with peptide in the GLP-1 bound structure, but with water in the cryo-EM structures of the GLP-1R bound to PF 06882961 and CHU-128 (right) were also assessed. All data are the means + S.E.M of at least four independent experiments performed in duplicate.

Figure S5.

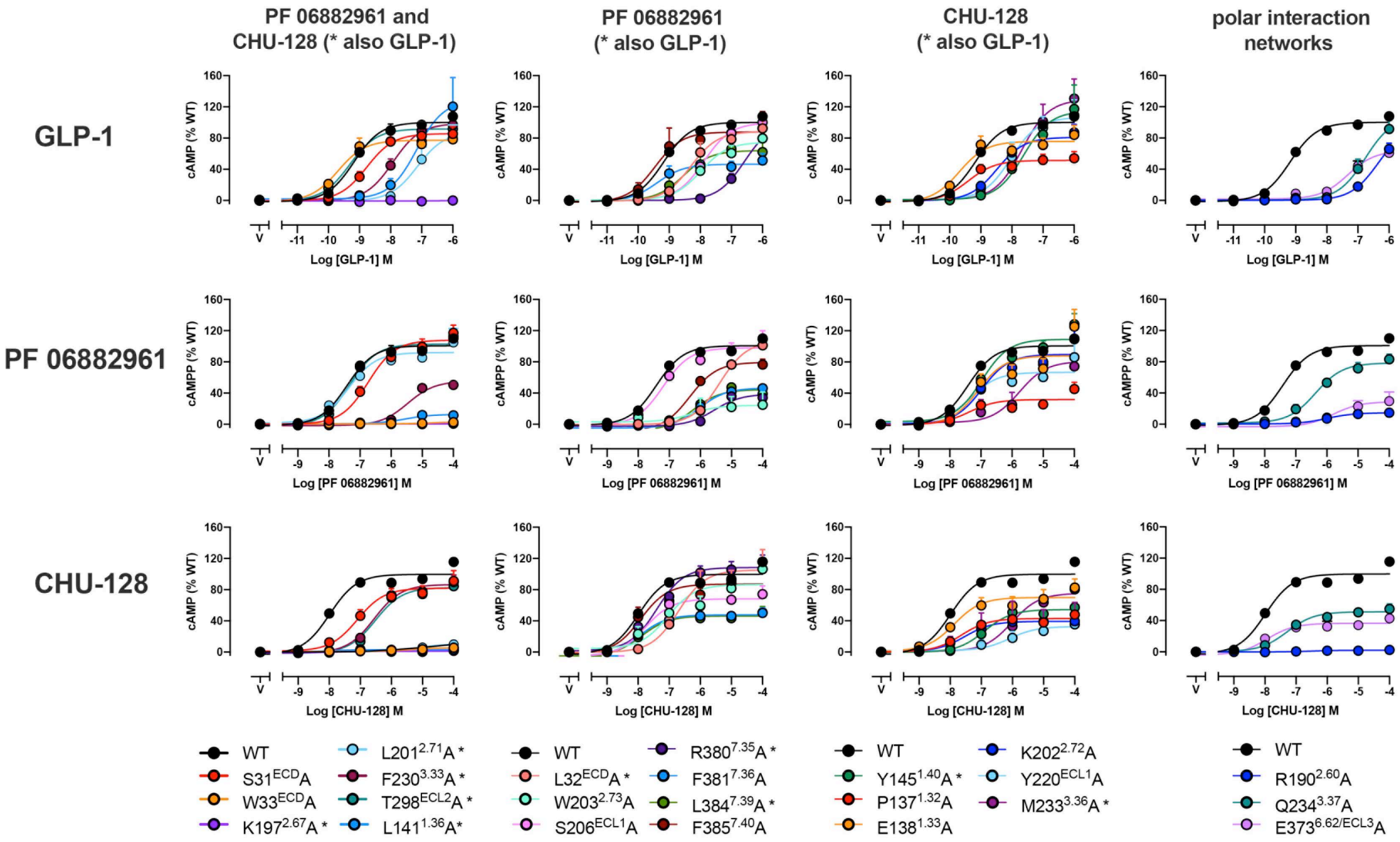

Mutated residue - Selected due to interactions in structures
PF 06882961 and CHU-128 (* also GLP-1)
PF 06882961 (* also GLP-1)
CHU-128 (* also GLP-1)
polar interaction networks
GLP-1
PF 06882961
CHU-128
cAMP (% WT)
cAMPP (% WT)
Log [GLP-1] M
Log [PF 06882961] M
Log [CHU-128] M
WT
S31ECDA
W33ECDA
K1972.67A*
L2012.71A*
F2303.33A*
T298ECL2A*
L1411.36A*
WT
L32ECDA*
W2032.73A
S206ECL1A
R3807.35A*
F3817.36A
L3847.39A*
F3857.40A
WT
Y1451.40A*
P1371.32A
E1381.33A
K2022.72A
Y220ECL1A
M2333.36A*
WT
R1902.60A
Q2343.37A
E373 6.62/ECL3A

Figure S6. The GLP-1R binding pocket shape, volume and properties differ in the presence of different agonists. *Related to Figures 4, 5 and Video S1.* **A,** Slice through of the upper portion of TM bundles bound to different agonists, coloured by hydrophobicity (top) and electrostatic potential (bottom). **B,** Overlay of CHU-128-bound structure with 6ORV (TT-OAD2 bound GLP-1R structure) revealed significant overlap in the ligand binding poses and commonalities in the conformation of TM6 and 7. **C,** Overlay of GLP-1-bound (orange), ExP5 bound (6B3J – yellow), CHU-128-bound (pale blue) and TT-OAD2 bound (6ORV – cornflower blue) GLP-1R structures shows differences in the TM6-ECL3-TM7 conformation.

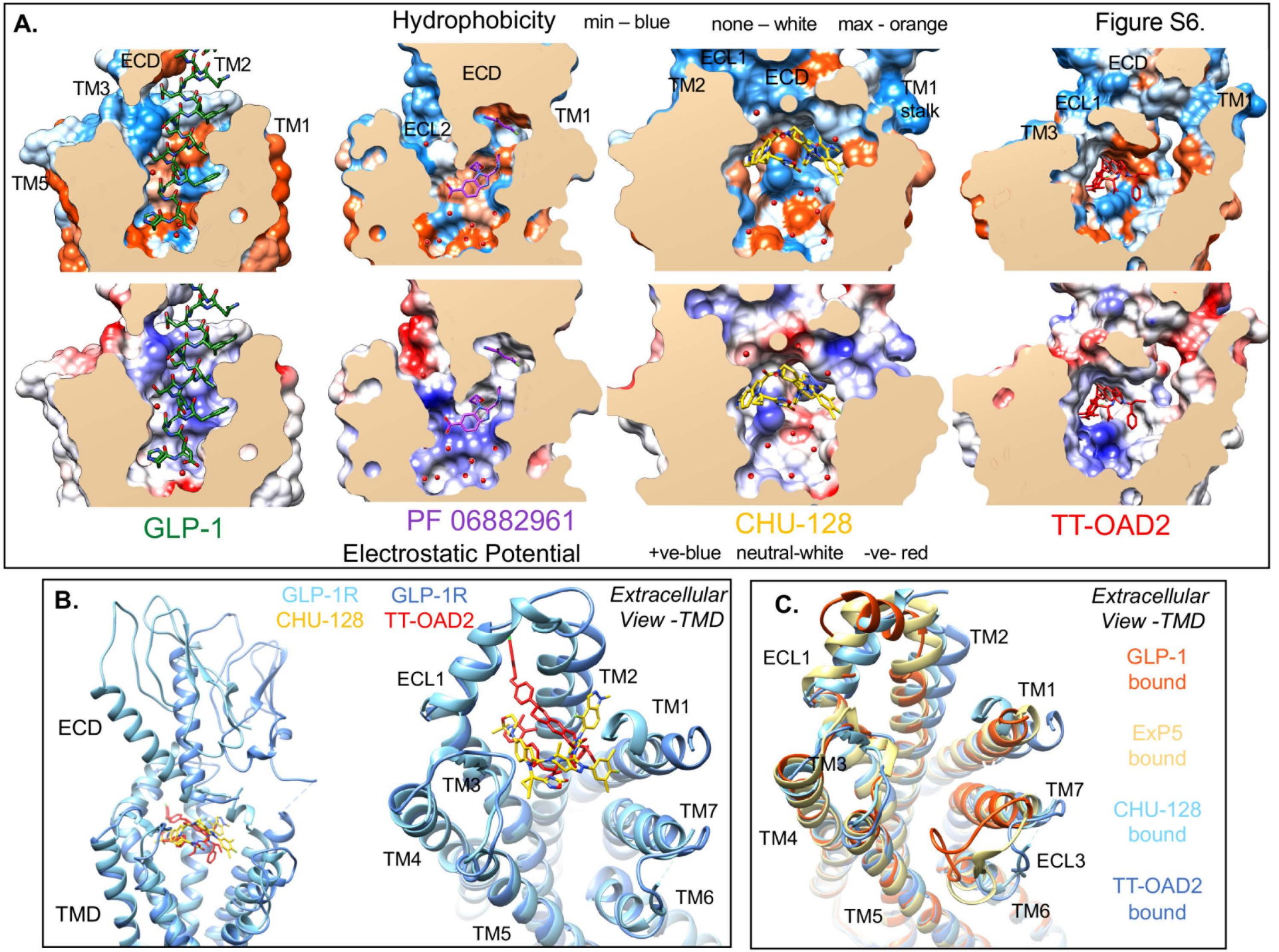
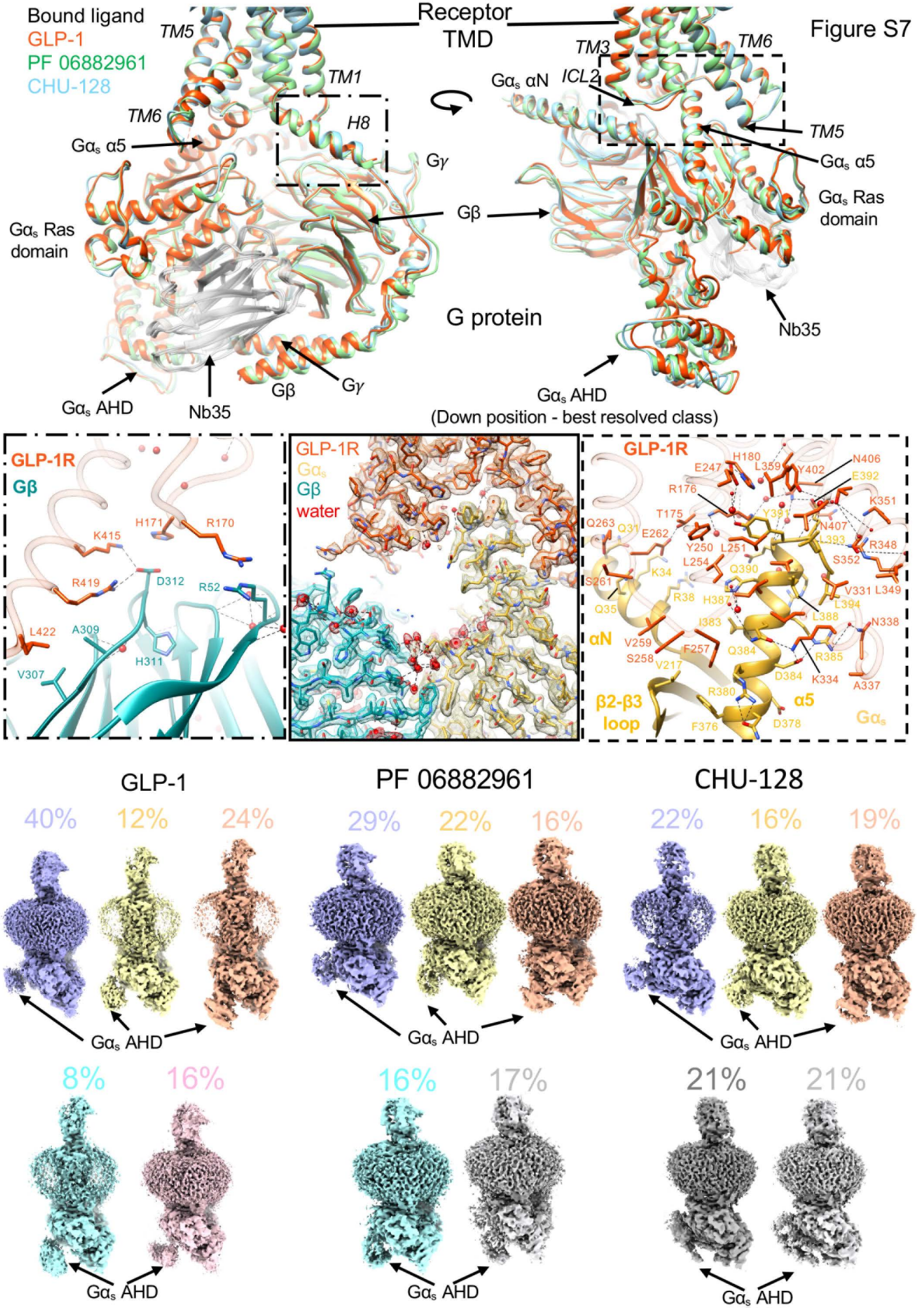

A.
Hydrophobicity
min – blue
none – white
max - orange
Figure S6.
ECD
TM2
TM3
TM1
TM5
ECD
TM1
ECL2
ECL1
TM2
ECD
TM1
stalk
ECD
ECL1
TM1
TM3
GLP-1
PF 06882961
CHU-128
TT-OAD2
Electrostatic Potential
+ve-blue
neutral-white
-ve- red
B.
GLP-1R
CHU-128
GLP-1R
TT-OAD2
Extracellular
View -TMD
ECD
TMD
ECL1
TM2
TM1
TM3
TM7
TM4
TM6
TM5
C.
Extracellular
View -TMD
ECL1
TM2
TM1
TM3
TM7
TM4
ECL3
TM5
TM6
GLP-1
bound
ExP5
bound
CHU-128
bound
TT-OAD2
bound

Figure S7. The GLP-1R:G_s complexes bound by different agonists share common interactions with G_s, yet differences in the mobility of the $G\alpha_s$ AHD. *Related to Figures 4 and 5 and Videos S3 and S4.* **Top,** superimposition of the GLP-1-bound (orange), PF 06882961-bound (pale green) and CHU-128-bound (pale blue) GLP-1R:G_s complexes reveal identical backbone conformations within the intracellular portion of the TM bundle, the G protein and the G protein: receptor interface. Rotamer positions of side chains in the lower portion of the receptor bundle were also conserved in the three structures (not shown). **Middle**; Interactions formed between the GLP-1R (orange) and the G protein (α - gold (right), β-cyan (left)) in the GLP-1 bound GLP-1R structure. Identical interactions were observed in the other two structures, including water (red spheres) mediated interactions. Middle panel shows a cut through of the model with the cryo-EM density showing the G protein:receptor interface. **Bottom,** 3D classification using a mask of the $G\alpha_s$, including AHD, revealed multiple conformations of this domain relative to the Ras domain. 3 common classes were observed in all three GLP-1R complexes (purple – "open/up", yellow – "mid" and coral – "closed/down", however, the % of particles in each class differed depending on the bound ligand; these are shown above the 3D classes. The other two classes differed between complexes and had less well-defined AHDs. The total number of particles used for the 3D classification were 615K, 732K, 933K for GLP-1, PF 06882961 and CHU-128, respectively. While not the most populated, the down conformation (coral) was the only class of sufficient resolution to enable modelling of the AHD and this is shown in panel A.


Figure S7
Bound ligand
GLP-1
PF 06882961
CHU-128
Receptor TMD
TM5
TM1
TM6
H8
Gαs α5
Gγ
Gβ
Gαs Ras domain
Gαs AHD
Nb35
Gβ
Gγ
TM3
TM6
Gαs αN
ICL2
TM5
Gαs α5
Gαs Ras domain
G protein
Nb35
Gαs AHD
(Down position - best resolved class)
GLP-1R
Gβ
H171
R170
K415
R419
D312
R52
L422
A309
H311
V307
GLP-1R
Gαs
Gβ
water
GLP-1R
H180
E247
L359
Y402
N406
E392
R176
K351
T175
Y391
Q263
Q31
E262
N407
L393
R348
Y250
L251
S352
L254
Q390
K34
S261
V331
L349
R38
H387
L394
Q35
I383
L388
N338
αN
V259
F257
Q384
R385
S258
V217
D384
K334
A337
R380
α5
β2-β3 loop
F376
D378
Gαs
GLP-1
PF 06882961
CHU-128
40%
12%
24%
29%
22%
16%
22%
16%
19%
Gαs AHD
Gαs AHD
Gαs AHD
8%
16%
16%
17%
21%
21%
Gαs AHD
Gαs AHD
Gαs AHD